

scenario) and with minimal operational efforts. For more details, refer to the “Further Reading” list at the end of this chapter.

BGP FRR

BGP fast reroute (FRR) accelerates BGP convergence time significantly to minimize packet loss and to avoid traffic black-holing situations after a failure event when IGP is tuned to converge in a subsecond, without sacrificing network stability. The BGP FRR concept enables BGP to use alternate paths within a subsecond following a failure event of the current active path. BGP FRR is achieved through BGP prefix-independent convergence (PIC) concept, where BGP installs a backup path or paths in RIB/FIB tables that are ready for immediate use. Therefore, a more predictable and constant convergence time of BGP can be achieved no matter how large the number of BGP prefixes is.

For example, in the MPLS VPN scenario in Figure 9-15, PE1 always considers PE2 as the preferred MP-iBGP next-hop to reach AS 600 prefixes. However, with BGP PIC, PE1 also installs an alternate path in its RIB/FIB tables (PE4 in this example). When PE2 fails, PE1 will be aware of this event through the removal of the host prefix by IGP in subsecond (assuming IGP is tuned to converge in subsecond) and will start to redirect traffic toward the alternate next-hop (via PE4). In the meantime the standard BGP convergence can take place while BGP PIC is forwarding the traffic through PE4 without introducing packet loss (or it will be very minimal, if there is any). As a result, BGP PIC offers more structured RIB/FIB tables (at the core or edge of the network) that helps network designers to align with the underlying IGP fast convergence as well as minimize packet loss and network instability following any network component failure.

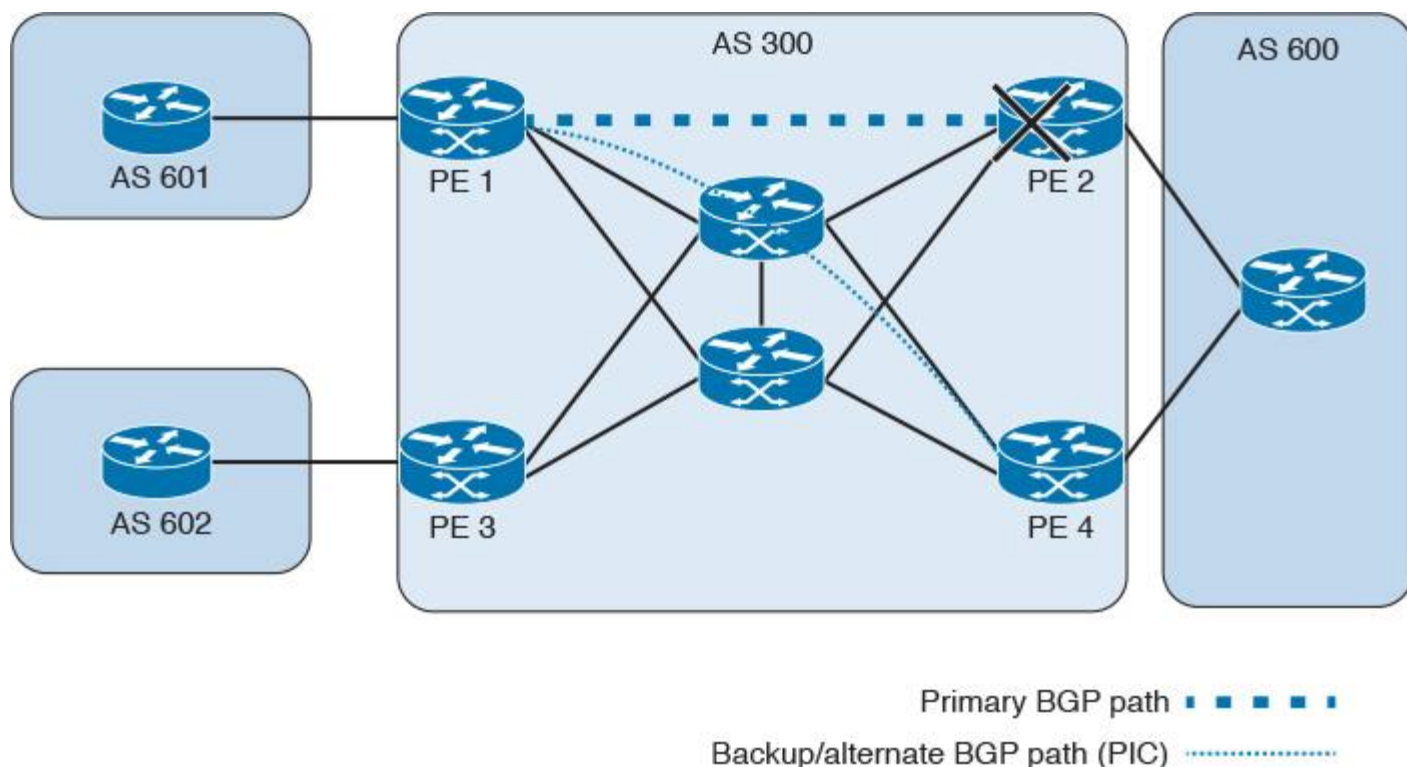


Figure 9-15 BGP FRR (PIC)

Similarly, BGP PIC offers more structured RIB/FIB tables that helps to achieve faster network convergence and minimize packets loss at the edge of the network, usually in an SP-style environment following a PE-CE link failure for multihomed CE sites. For instance, AS 600 in Figure 9-16 can be protected with BGP PIC to achieve BGP FRR using the same concept discussed earlier, where PE2 holds primary and backup paths (primary via the directly connected PE-CE link and the backup via PE4 best external). As soon as PE2 detects the link to AS 600 CE failure, traffic will be shunted to PE4 to reach the AD 600 CE to avoid packet loss, while BGP will continue its typical route withdrawal and update propagation to converge the network, as illustrated in Figure 9-16. However, because PE-CE communication is normally over an external BGP (eBGP) session (no IGP to react to the failure and updated RIB/FIB), bidirectional forwarding detection (BFD) is

required to detect a failure of an eBGP one-hop peer in less than a second. This can be combined with the BGP failover feature to track the availability of the BGP neighbor (CE), not only based on link status.

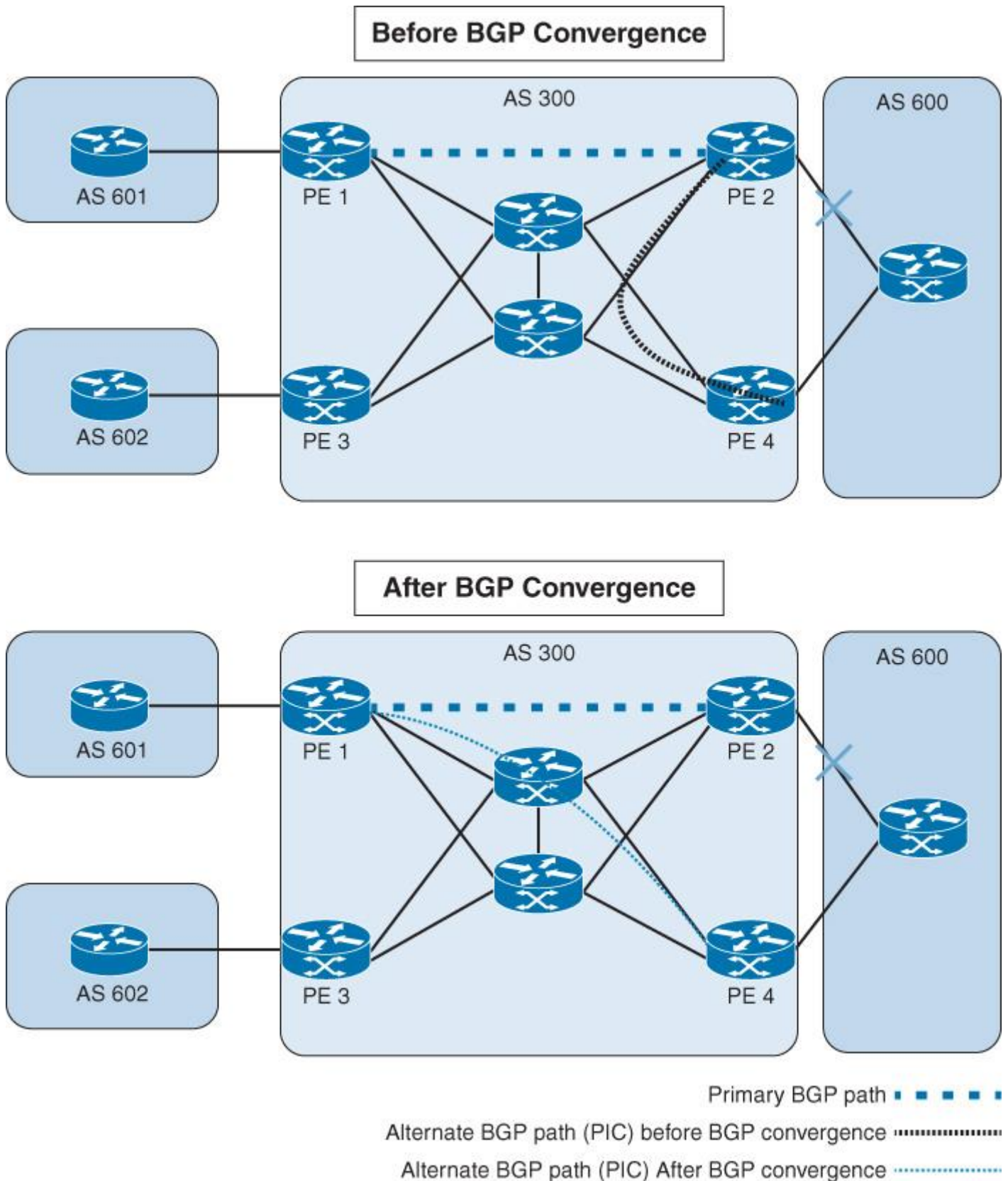


Figure 9-16 BGP FRR (PIC Best External)

Note

Typically, at the FIB level, after a failure event where a next hop to a prefix fails, the FIB (Cisco CEF) will look for another path to reach the same prefix by searching through the FIB table to find the next longest matching path to the prefix. This proves beneficial when the next hop is several hops away and there are multiple paths to reaching that next hop, such as a router with a loopback interface IP address that is reachable via multiple physical interfaces/paths. However,

this feature (recursion) may impact and conflict with the BGP PIC concept, because there is a precomputed alternate path already in place with BGP PIC. Moreover, the searching process through the FIB table almost always slows down convergence time in general. Therefore, ideally, the CEF recursion feature should be disabled at least for the learned host routes when BGP PIC is in use. In some Cisco platforms, “When the BGP PIC functionality is enabled, Cisco Express Forwarding recursion is disabled by default for next hops learned with /32 host routes and next hops that are directly connected.”⁴

4. “BGP PIC Edge for IP and MPLS-VPN,” <http://www.cisco.com>

BGP Multiple Path Announcement Solutions

The natural behavior of BGP, as defined in RFC 4271, is to announce only a single path to its BGP neighbors, known as the *BGP best path*. However, in some scenarios, the announcement of multiple paths within BGP is becoming a requirement, especially when BGP fast convergence is required. Furthermore, BGP path hiding of BGP RRs may prevent the efficient use of BGP multipath or keep BGP PIC from working properly. BGP path hiding also inhibits fast and local recovery because the network has to wait for the standard BGP control plane convergence to restore traffic across the network [34]. Therefore, a solution that offers path diversity is required for networks that need to use the available path or need to maintain a fast network convergence at the control plane level (BGP). Multiple solutions are available to change this default behavior of BGP and facilitate the ability to advertise multiple paths with BGP, some of which were already discussed in Chapter 5 with regard to hot-potato routing with RR. In addition, these mechanisms serve to optimize BGP FRR in similar scenarios:

- BGP additional paths
- BGP best external
- BGP diverse paths
- Different route distinguisher (RD) approach with MP-BGP in MPLS VPN environments

Summary

This chapter covered high-availability network design and the primary elements and approaches to achieve it at multiple layers. This chapter also discussed how the concept of network convergence consists of multiple elements that all collectively work together to achieve the overall desired convergence time. In addition, two primary approaches at the protocol level (the network protection approach and the network recovery approach) to achieve the desired level of network resiliency were also examined, with each having advantages and limitations, as summarized in Table 9-4.

	Network Protection Approach	Network Recovery Approach
Scalability	Moderate	High
Convergence	Very fast	IGP potentially fast with tuning
Hardware resource consumption (memory, CPU)	Moderate to high (depends on the number of protected routes/LSPs)	Moderate (depends on some factors such as number of routes and routing domains design)
Design and configuration complexity	Protocol dependent (for example, LFA simpler than MPLS-TE FRR)	Simple
Operational complexity	The larger, the higher	Moderate (however, more prone to human error [such as mismatching timers])
Network stability with subsecond convergence	Stable (taking into consideration the fail-over time and frequency between the protected and back paths)	High possibility of instability and microloops
Solution example	LFA, MPLS-TE FRR	Tuned IGP for fast convergence

Table 9-4 Network Protection Approach Versus Network Recovery Approach

Last but not least, this chapter explained, in multiple places, that the level of availability will vary from network to network and from part to part within the same network, based on multiple variables and mainly driven by the application criticality and business priorities. In other words, not every network or every part of the network needs HA considerations or fast network convergence. For network designers to identify when and where to consider it, a good understanding of the application, functional, and business requirements is required. (For instance, identify what is critical and high priority to the business, versus what is important but not high priority, versus what is not critical and not high priority.)

Further Reading

“High Availability Campus Recovery Analysis Design Guide,” <http://www.cisco.com>

RFC 6571, *Loop-Free Alternate (LFA) Applicability in Service Provider (SP) Networks*, <http://www.ietf.org>

“Remote Loop-Free Alternate (LFA) Fast Re-Route (FRR),” draft-ietf-rtgwg-remote-lfa-11, <http://www.ietf.org>

RFC 6413, *Benchmarking Methodology for Link-State IGP Data-Plane Route Convergence*, <http://www.ietf.org>

RFC 4090, *Fast Reroute Extensions to RSVP-TE for LSP Tunnels*, <http://www.ietf.org>

“MPLS Traffic Engineering (TE) - Autotunnel Primary and Backup,” <http://www.cisco.com>

Part VI: Other Network Technologies and Services

Chapter 10 Design of Other Network Technologies and Services

Chapter 10. Design of Other Network Technologies and Services

This chapter covers multiple networking and IP service design concepts and considerations that are considered as noncore topics for the purpose of the CCDE practical exam at the time of this writing. Although these topics are not considered as core technologies for the CCDE exam, one or more of these technologies are expected to be part of each scenario. The different topics discussed in this chapter might be presented as an application or service to be used to achieve a business need. For example, a business-critical application might require multicast to be enabled across the network to work properly.

Note, as well, that this chapter focuses on the design drivers, considerations, and approaches that network designers can consider based on the different design requirements, without covering any deep technical details. This chapter covers the following:

- IPv6 design considerations
- IP multicast design considerations
- Quality of service (QoS) design considerations
- Network security design considerations
- Network management

IPv6 Design Considerations

IPv6 is a broad topic. Therefore, this section focuses on one of the most critical and important topics, which is the integration and coexistence of IPv4 and IPv6, by covering the different design and technical options and how to follow a business-driven lifecycle design approach.

IPv6 Business and Technical Drivers

Although the initial IPv6 draft “The Case for IPv6” was published as an IETF draft more than a decade ago (1999) (draft-ietf-iab-case-for-ipv6-06.txt), there was no serious move or adoption of IPv6 by enterprises and service providers. However, during the past few years, billions of smartphones and various mobile devices have been connecting to the public Internet and private networks. The overall Internet needs to evolve and grow to support this continuous growth, not to mention that the increased adoption of Internet-based services and public cloud hosting has added more demand on public IP addresses.

Furthermore, the new trends and technologies, such as the Internet of Things (IoT) and smart connected cities, are leading to an increasingly growing number of Internet-connected users, applications, sensors, and services that require a huge number of IPs. According to Gartner, Inc., “The Internet of Things (IoT), which excludes PCs, tablets, and smartphones, will grow to 26 billion units installed in 2020, representing an almost 30-fold increase from 0.9 billion in 2009” [92]. As the Internet transitions, organizations must start considering adopting IPv6 to support and facilitate achieving their business goals in terms of organic growth, merger and acquisition, and continuity in the market. With the exhaustion of IPv4 addresses and the increased demand on IPs, organizations face many challenges, such as the following:

- The exhaustion and constraints of IPs (IPv4), which adds many challenges to managing and provisioning new services that require new addressing spaces

■ Added complexity to merger and acquisition scenarios, where Network Address Translation (NAT), with its limitations to overcoming the overlapping addressing issue, is considered a primary design option

■ New market and service trends, where mobility and numerous connected devices (smart cities, IoT, and so on) are key drivers for both enterprise and service providers that are trying to adopt innovation and the provision of new services, for which a large number of IPs are required to accommodate the various IP-enabled endpoints

Therefore, the adoption of IPv6 is becoming a necessity to overcome these challenges and to meet these new technology solution requirements and demands. Hypothetically, if an organization is encountering one or more of the following situations, it is time to migrate or at least start to consider enabling IPv6:

■ Unable to expand the business to other global regions because of the exhaustion of the public IPv4 addresses.

■ Deploying innovative network environments, applications, and devices such as IoT with a huge number of connected sensors required to support smart connected communities.

■ A service provider or an enterprise needs to maintain seamless connectivity across fixed and mobile users when using collaborative applications, and NAT is no longer an option.

■ Enabled IPv6-based 4G/LTE mobile networks or connecting workforces to these networks.

■ Organizations that primarily work as a supplier or partner with public sector or government entities are required to comply with their standards. In these entities, IPv6 is fast becoming the standard.

■ Encountering IPv6 need in the network driven by the deployments of end-user and host operating systems and applications such as Microsoft Windows 7, Windows 10, Windows Server 20xx, Apple OS X, system virtualization, and large scale multitenancy.

IPv6 Addressing Types (Review)

Before taking a look at the different types of IPv6, it is important that network architects and designers be aware of the high-level differences and similarities between the two IP versions. [Table 10-1](#) summarizes some of the technical similarities and differences between IPv4 and IPv6.

	IPv4	IPv6
Address scope	32 bit	128 bit, multiple scopes
IP allocation	Manual, Dynamic Host Configuration Protocol (DHCP)	Manual, stateless address autoconfiguration (SLAAC), DHCP
Address types	Unicast, multicast, broadcast	Unicast and multicast
QoS	Differentiated services, integrated services	Differentiated services, integrated services, flow label
Multicast	Internet Group Management Protocol (IGMP), Protocol-Independent Multicast (PIM), Multiprotocol Border Gateway Protocol (MP-BGP)	Multicast Listener Discovery (MLD), PIM, MP-BGP
Security	No built-in- support	IPsec built in

Table 10-1 Summary of IPv4 Versus IPv6

Note

Although IPv6 supports built-in IPsec, it is considered a myth and misconception that IPv6 is more secure than IPv4. This assertion stems from the original mandated use of IPsec in host-to-host communication, as specified in RFC 2401. Consequently, if IPsec is implemented, it will provide confidentiality and integrity between two hosts, but it still will not address any link operation vulnerabilities, attacks, and most of the denial-of-service (DoS) attacks.¹

1. http://www.cisco.com/c/dam/en/us/products/collateral/ios-nx-os-software/enterprise-ipv6-solution/aag_c45-707354.pdf

With regard to IPV6 addressing, IPv6 has three types of unicast addresses:

■ **Link local:** This address is nonroutable and can exist only on a single Layer 2 domain (fe80::/64). As described in RFC 4291, even when one or more routable IPv6 addresses are assigned to a certain interface, an IPv6 link-local address is still required to be enabled on this interface.

■ **Unique local address (ULA):** This address, also known as a *site-local address*, is routable within the administrative domain of a given network (fc00::/7). This address, in concept, is similar to the IPv4 private address range (RFC 1918).

■ **Global:** This address is routable across the Internet (2000::/3). This address is similar in concept to the IPv4 public ranges.

Network designers can mix and match the different IPv6 addressing types when designing an enterprise network. However, there are some issues associated with each design model, as summarized in [Table 10-2](#).

IPv6 Addressing Model	Scope	Design Simplicity	Manageability	Scalability
ULA only	Internal	Moderate	Moderate	Low
ULA + global	Internal and external	Complex	Complex	Moderate
Global only*	Internal and external	Simple	Simple	High

* Because NAT is primarily used with IPv4 to overcome IP address shortages, it is always arguable that this is a less-secure model because it exposes internal device IPs to external networks (no address hiding using NAT). In addition, it is commonly claimed that NAT offers a layer of secure communication; however, that argument fails to cover the impact of NAT on many applications at higher layers. “This is amplified through the widespread sharing of vendor best practice documents and sample configurations that do not differentiate the translation function of address expansion from the state function of limiting connectivity.”²

Table 10-2 IPv6 Address Types²

2. Local Network Protection for IPv6, RFC 4864

Migration and Integration of IPv4 and IPv6

Considerations related to transitioning to IPv6 and planning and designing the integration of both IP versions to coexist are critical aspects since the introduction of IPv6. However, the most common critical and challenging question for network architects and designers with regard to migrating and enabling IPv6 is where to start. For network architects or designers to achieve a successful IPv6 migration or integration, they must follow a structured approach based on the top-down design methodology, which consists of network discovery, assessment, planning, design, implementation, monitoring, and optimization, as shown in [Figure 10-1](#).

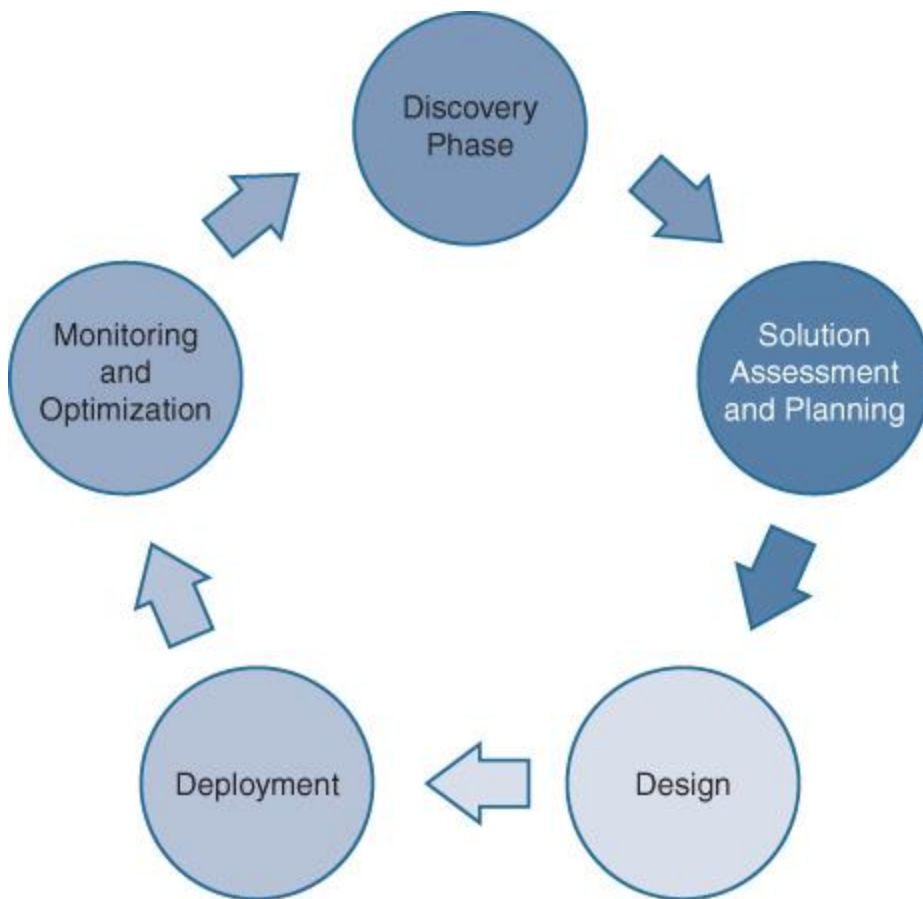


Figure 10-1 *Migration to IPv6 Design Approach*

The sections that follow explain the phases shown in [Figure 10-1](#).

Discovery Phase

At this phase, network architects or designers usually focus on understanding and identifying the business goals and drivers toward the enablement of IPv6, in addition to other influencing factors such as project timeframe, government compliance, and the geographic distribution of the sites with regard to IP addressing availability. Information about other influencing factors from the network point of view at this stage also needs to be identified and gathered at a high level, such as whether the existing network infrastructures (LAN, WAN, security nodes, services, and applications) support IPv6 and whether the business is willing to invest and upgrade the nodes that do not support IPv6. Therefore, it is critical that the right and relevant information be gathered at this phase so that it can be analyzed and considered during the planning phase.

Solution Assessment and Planning

Migrating or integrating IPv4 and IPv6 networks might seem as simple as enabling IPv6 routing across certain network areas; however, the reality is that several other factors must be considered during the planning phase to ensure that the design will deliver its intended goal and be prepared for the challenges and constraints that may arise during the deployment phase. After the completion of the high-level business and network discovery, network designers need to analyze each of the identified influencing factors and generate a migration or integration plan based on the gathered information. In general, the following considerations help at this phase to influence and drive the detailed design of the transition strategy and approaches at later stages.

■ **Goal:** Understanding the main purpose of the migration or integration (based on the information gathered during the previous phase) is important to driving the design in the most suitable direction. For instance, is the transition to IPv6 to access certain services in the data center? To comply with “regularity compliance”? Or to enable IPv6 at the Internet edge only (for example, lack of public IPv4 pools)?

■ **Infrastructure support:** Furthermore, the answers with regard to the goal become vital during the planning phase, because it is important to understand whether the entire infrastructure supports IPv6 or only the network edges (for example, provider edges [PEs] in service provider [SP] network and access or distribution in campus networks) and whether the business is willing to upgrade the devices that do not support IPv6. Based on this, the migration or integration plan guides the next phase (design) as you consider a technology solution to overcome any constraint with regard to IPv6 support across the network.

■ **Existing services and applications:** Although many applications already support IPv6, many still do not, especially those that are developed in house. Therefore, understanding and assessing application support is critical because the upgrade, if it fails to consider how the IPv6 networks will reach these IPv4-only applications, will break the communication with these applications. Such a break may seriously impact the business.

In other words, the planning phase takes the output information from the discovery phase and analyzes it and uses it as a foundation to drive the selection of the appropriate migration/integration approach. Consider, for example, that an enterprise needs to migrate its network to be IPv6 enabled (end to end), but at present the core network components do not support IPv6 and the business is not allocating any budget to upgrade these components. In addition, access to some new IPv6-enabled applications hosted at the data center is an urgent requirement. This information should be collected during the discovery phase, and the network designer at the assessment and planning phase is expected to select the right approach to meet the requirements for this enterprise, taking into consideration the relevant constraints. In this case, the network designer may suggest either an IPv6 over IPv4 tunneling mechanism (either sourced from the workstation or using access/distribution switches), Domain Name System (DNS)-based translation, or 64NATing to facilitate accessing the new IPv6-based applications over the existing IPv4 core infrastructure.

The following subsections cover the approaches possible during the planning phase with regard to IPv6 based on the targeted environment (enterprise versus service provider). The selected approach should ideally be complemented with one or more of the technical mechanisms listed later in [Table 10-5](#) at the design phase. Therefore, each phase (ideally) must take the outcomes of the previous phase as a foundational input to achieve a cohesive end-to-end business-driven design that avoids any “design in isolation” throughout all the phases.

Transition Approaches to IPv6 (Enterprise Networks)

After identifying the objectives of the migration or integration of IPv4 and IPv6, network designers must decide which approach is the most suitable, taking into consideration the objectives, the different requirements, and design constraints, such as timeframe, available budget, and whether IPv6 is supported by the network nodes. [Table 10-3](#) summarizes the most common approaches, based on different design goals and priorities, to transition an enterprise network to be IPv6 enabled.

Design Goal	Priorities	Timeframe	Design Approach	Design Considerations
Migrate the enterprise network to be pure IPv6 or dual stack	No service interruption	Flexible	Migrating the core to be in dual-stack mode first, then other enterprise modules can be gradually migrated to IPv6. (Either only IPv6 or dual stack depends on the goals and requirements.)	Increased utilization of hardware resources. Increased control plane complexity in dual-stack mode. Core network component must support IPv6. Partially migrated blocks may require tunneling as an interim solution.
Migrate the enterprise network (fully or partially) to be pure IPv6 or dual stack	Certain enterprise modules have to be (quickly) migrated first, such as data center	Limited	Migrating certain modules of the enterprise network first. DNS translation or tunneling mechanism such as ISATAP is required to maintain the communications between IPv6 and IPv4 islands within the network.	This approach is suitable when the core device does not support IPv6 and requires either hardware or software upgrades. Increases design and control plane complexity. Increases operational complexity.

Migrate the data center network (fully or partially) to support communication with IPv6 hosts	Support virtualized and nonvirtualized IPv6 hosts	Flexible	Depends on the data center architecture different approach can be used. For example, a DC design can consider a typical dual stack, or an overlay can be considered using VxLAN to support IPv6 hosts; also MPLS enabled DC may consider 6PE/6VPE.	In all cases, dual stack increases hardware resource utilization, control plane complexity, and operational complexity. If the timeframe is limited, one quick and common solutions is host-based tunneling (ISATAP, RFC 5214), assuming endpoints and hosts support this tunneling mechanism.
Provide IPv6 access either inbound or outbound at the enterprise Internet edge	Support translation between IP4 and IPv6 at the Internet edge	Flexible	A translation mechanism is required that is either based on a load balancer, pure DNS, or classical “64” NATing based.	Increases operational complexity at the Internet edge. Requires additional security consideration with regard to IPv6 enablement. If the timeframe is limited, it is always quicker to consider the current infrastructure support. (For example using NAT at the existing Internet edge router is quicker than provisioning a new DNS server to do the translation.)

Table 10-3 Approaches to Enable/Transition to IPv6: Enterprise

Transition Approaches to IPv6 (SP Networks)

In its overall approach and goal, enabling IPv6 in an SP network is slightly different from in enterprise networks. Typically, the SP networks enable IPv6 either to provide a transit path for other SPs (such as transit and peering Internet service providers) or to offer IPv6 connectivity to its customers. Which mechanism to use is mainly driven by the goal and the transport used by the SP, whether it is based on native IPv4 only or Multiprotocol Label Switching (MPLS) based (with or without MPLS VPN). [Table 10-4](#) summarizes these approaches from the SP point of view.

Goal	Transport	Possible Approaches
Provide IPv6 Internet transit	Native IPv4	Dual stack, tunneling (manual RFC 2893, generic routing encapsulation [GRE], Layer 2 Tunneling Protocol Version 3 [L2TPv3])
Provide IPv6-based services and Internet access to residential clients	Native IPv4	Dual stack, 6rd, tunneling such as IPv6 over L2TP
Provide IPv6 Internet access/transit	MPLS	6PE, IPv6 over pseudowires
Provide IPv6 connectivity for MPLS L3VPN customers	MPLS	6VPE
Provide IPv6 Internet access for MPLS L3VPN customers over a separate virtual private network (VPN) or within the same customer VPN	MPLS	6VPE

Table 10-4 Approaches to Enable/Transition to IPv6: SP

Unlike with enterprise networks, enabling IPv6 in an SP network can be more flexible and less interruptive, as covered in [Chapter 5](#), “[Service Provider Network Architecture Design](#).” SP networks by nature are transport networks where no endpoints or hosts are directly connected. This means that there is no need to consider any directly connected host with this transition, except host and services used by the SP to deliver specialized functions such as network management, DHCP, and authentication/authorization services. Besides, when MPLS is enabled (nowadays MPLS to a large extent is the de facto protocol for SP networks), the integration and enablement of IPv6 will be simpler with the MP-BGP overlay capability (MP-BGP-based 6PE, 6VPE). Furthermore, operators can consider the phased approach in which only the PE nodes that need to provide IPv6 transit connectivity need to be enabled with IPv6 first without introducing any change to the core (P) routers. However, the additional load and hardware resource limits must be calculated and taken into consideration.

Note

One of the primary considerations that must be taken into account when migrating or integrating with IPv6 is to secure IPv6 in the same manner that IPv4 is secured. Otherwise, the entire network will be vulnerable to network attackers and breaches. For instance, since the release of Microsoft Windows Server 2008, IPv6 has been native to the Windows OS, which supports transition technologies at the server/client level such as ISATAP. In this case, if one of these servers is compromised and the network security rules do not consider IPv6, malicious traffic can ride an IPv6 tunnel or packet without being blocked or contained by the security devices in the path.

As covered so far, the migration to IPv6 differs slightly in SP networks as compared to in enterprise networks, based on the fact that SPs mainly need to provide a transit path to their customers and other peering providers. However, today’s SPs are not limited to just providing transit paths; they also offer many other services, such as hosted services and applications, software as a service (SaaS), cloud-based hosting data

centers, and content services such as IPTV. Practically, service providers cannot simply migrate to native IPv6, because they normally have other services and customers who need IPv4. Therefore, the coexistence of IPv4 and IPv6 is inevitable nowadays for SPs. What is important to plan and design for is the strategy that can be used to incorporate IPv6 into the existing provider network in a way that is not interruptive and at the same time offers enough flexibility and reliability to optimize SPs' time to market with regard to IPv6-enabled services and transport while maintaining IPv4 support intact. One common approach that application and content provider use to support IPv6 is to enable IPv6 at the services level first. With this approach, providers that offer hosted services and applications, whether it is cloud offering (such as SaaS) or any other IP service such as VoIP and Internet, usually enable or migrate these services to be IPv6 enabled first, and so customers that need to use these services must be IPv6 enabled, using a form of translation at the customer side such as NAT v4 to v6 at the enterprise Internet edge, or the provider might offer the translation service at a different cost or as value-added service such as DNS-based translation. Also, the SP must ensure the underlay transport can carry IPv6 communications. In fact, this approach is similar (to a large extent) to enabling or rolling out IPv6 applications in an enterprise data center in which the enterprise needs to incorporate an interim solution to provide and maintain access to these applications to users with only IPv4 endpoints.

When the desired approach is decided on based on the design requirements and constraints, network design can move to the next phase to start connecting the dots and generate a solution driven by the outcomes of the previous phases.

Detailed Design

After selecting the suitable approach for the migration or integration between IPv4 and IPv6 networks, ideally based on the gathered and analyzed information during the planning phase, network designers at this stage can put together the details of the design, such as selecting the suitable integration mechanism, deployment details such as tunnels termination, IP addressing, routing design, network security details, and network virtualization considerations, if any are required. Typically, the outcome of the design phase will be used by the implementation engineers during the deployment phase to implement the designed solution. Therefore, if there is any point that is not doable or practically cannot be implemented, it will be reported back to this phase (to the network designer) to be revised and changed accordingly. There are various mechanisms and approaches with regard to integrating IPv6 and IPv4. For simplicity, these mechanisms can be classified as following:

- Dual stack
- Tunneling based
- Translation based
- MPLS environment solutions

Table 10-5 lists the various technical mechanisms that can be used to integrate and support the coexistence of both IPv4 and IPv6, taking into consideration some of the primary design aspects that influence the solution selection based on the design requirements.

Mechanism	Scenario	Targeted Environment	Design Concern
Dual stack	End-to-end IPv6 + IPv4.	Any environment that ultimately needs to move to end-to-end IPv6	IPv6 support in all L3 aware platforms is required. Increased control plane complexity. May introduce scalability weaknesses when both IP versions are running together (depends on the available hardware resources such as memory).
Tunneling: point to point (P2P) (L2TPv3, GRE RFC 2473)	Transit IPv6 over IPv4-only network.	Small number of IPv6 islands that need to interconnect over IPv4 network	Scalability, encapsulation overhead. Increased control plane complexity.
Tunneling: ISATAMP (RFC 5214)	Host-sourced tunnels that terminate at IPv6-enabled modules or services.	For trial IPv6 services or in case of IPv6 enabled partially (for example, only at the DC) (mostly enterprise networks)	Affects the overall network architecture. QoS, multicast, and NAT issues. Adds control plane complexity. Increases operational complexity.
Tunneling: mGRE	Interconnect IPv6 over IPv4 in a hub-and-spoke topology.	Interconnects IPv6-enabled remote sites in hub-and-spoke topology over IPv4 WAN Interconnects private IPv6 islands across public IPv4 cloud	Multicast traffic has to go via the hub. Adds control plane complexity. Increases operational complexity.

Tunneling: 6rd (RFC 5969)	Used to extend IPv6 deployment to customer sites (usually residential gateway), with limited impact on existing IPv4 infrastructure.	SP networks that offer IPv6 services/ Internet access over IPv4 SP network to residential customers Simple, stateless, automatic IPv6-in-IPv4 encaps/decap that offers fast IPv6 enablement	Whether the network equipment supports 6rd. Adds control plane complexity. Increases operational complexity.
Tunneling: IPv6 over L2TP software	To offer IPv6 access for residential gateway.	Digital subscriber line (DSL)/residential SPs With limited investment Stateful architecture on L2TP network server (LNS)	Dual-stack IPv4/IPv6 service on residential gateway LAN side. Increases operational complexity.
Translation 64: NAT/SLB	Allow IPv6 handsets/endpoints to access IPv4 Internet over Long Term Evolution (LTE)/3G or IPv4 services.	Green-field IPv6 service providers or enterprise networks that need to be interconnected to legacy/existing IPv4 network/services	Does not support every application type or protocol today Performance may not match dual-stack design, depending on traffic load.
Translation: DNS	DNS in this scenario offers the translation between the v4 and v6 based on the source and targeted host.	Access applications or services by name	Limited to services and applications that can be reach by name, not IP.

Translation: LISP	LISP encapsulation can facilitate the IPv6 communication only over IPv4 transport.	enterprise edge, DC or WAN with mix of IPv4 and IPv6 networks	High operational complexity. Increased control plane complexity. Network devices must support LISP. ³
MPLS: 6PE	Facilitate enabling IPv6 over an existing MPLS and MP-BGP IPv4 network.	Large enterprises and SPs that want to provide IPv6 over their IPv4 infrastructure	Does not provide traffic separation between different customer networks (no MPLS VPN support). May introduce scalability limitation because a separate Routing Information Base (RIB) and Forwarding Information Base (FIB) is required per customer Increased control plane complexity.
MPLS: 6VPE	Facilitate enabling IPv6 over an existing MPLS/MP-BGP IPv4 network for VPN customers.	MPLS VPN providers or enterprises with MPLS VPN networks	Increased control plane complexity.

Table 10-5 Mechanisms to Support Coexistence of IPv4 and IPv6³

3. “Enterprise IPv6 Transition Strategy Using the Locator/ID Separation Protocol,” <http://www.cisco.com>

Note

Information in [Table 10-5](#) is not a best practice or a must to be followed with regard to IPv6 integration and migration options. However, it is based on the most commonly considered technology solutions for certain scenarios. And network designers must always assess the different influencing factors before suggesting any approach or mechanism.

Note

Typically, adding any overlay or tunneling mechanism to the network will almost always increase the level of its operational complexity. However, this level varies based on several factors, such as network size, routing design, staff knowledge, and the nature of the selected technology itself.

Deployment, Monitoring, and Optimization

These steps usually are more relevant to the implementation of the design followed by continuous monitoring for the implemented solution to ensure that the network is delivering the promised value and

meeting the expectations. Ideally, the implementation should follow an implantation plan that specifies what the services are, and the futures that need to be enabled before proceeding with any step. It should also cover any potential risk associated with any change (for example, whether it is going to be interruptive to the existing production network). For instance, enabling IPv6 at the routing protocol level may lead to resetting the existing IPv4 peering sessions; normally this depends on the routing protocol, hardware platform, and software in use.

Transition to IPv6: Scenario

ABC Corp. is an international real-estate company headquartered in Singapore, with 116 remote sites distributed across Asia, Australia, and Europe, as shown in [Figure 10-2](#).

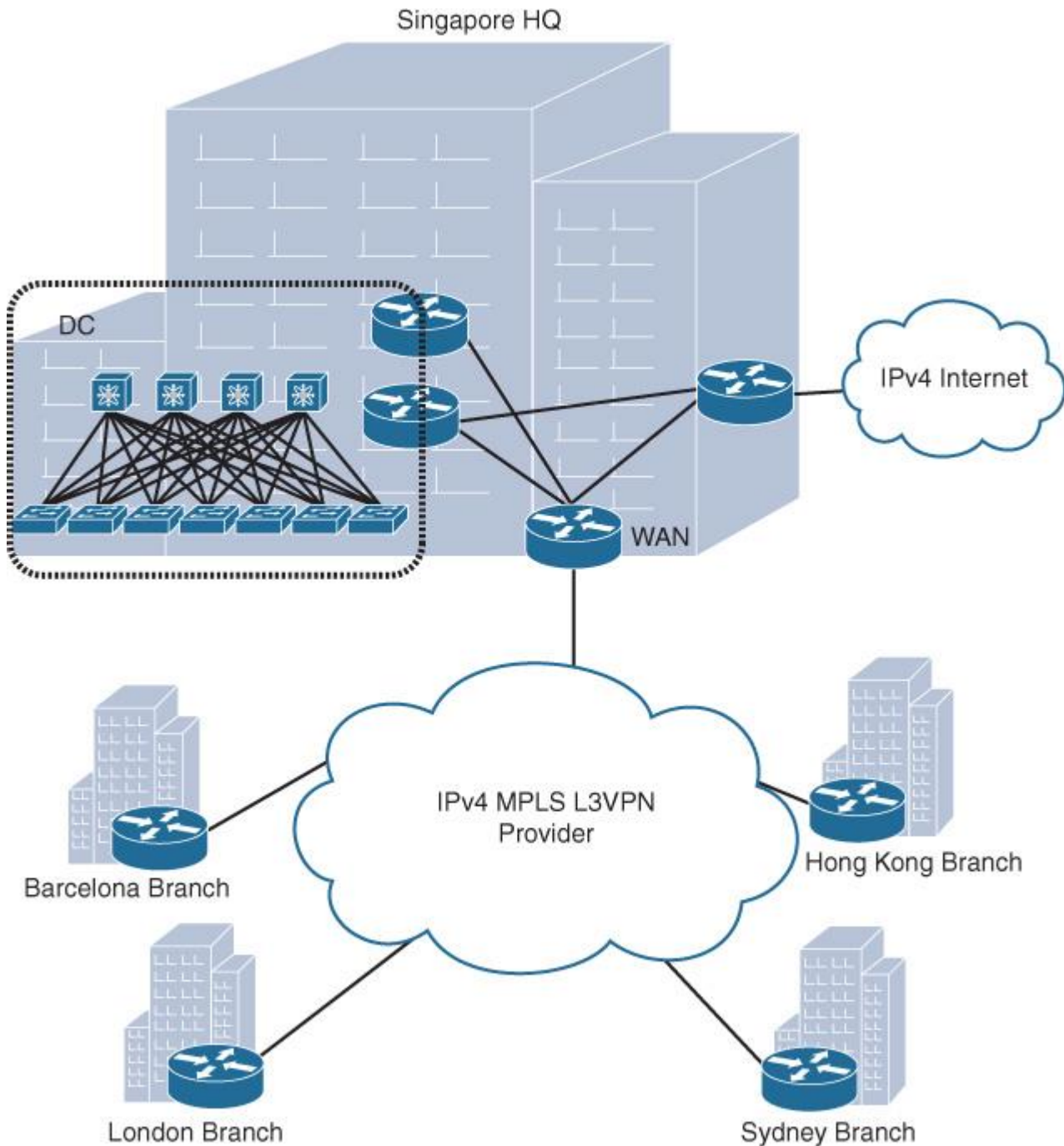


Figure 10-2 ABC Corp. Network

The CIO of ABC Corp. has decided to migrate their entire IP network infrastructure and applications to be primarily based on IPv6 to support some of the long-term business innovation and strategic plans. In addition, ABC Corp. wants to achieve the following with this transition project to maintain business continuity:

- Retain the ability for their internal users to access some legacy applications that do not support IPv6 and to access the IPv4 Internet

- Provide the ability for external users to access the ABC Corp. new IPv6 web-based services over the IPv4 Internet

Therefore, they decided to add another Internet link dedicated to accessing IPv6 Internet services. Moreover, the security team of ABC Corp. requires accessing some (pre-defined) IPv4 Internet websites, “web-based services,” by the internal IPv6-enabled users to appear as if they are accessible over the IPv6 Internet. For example, when an IPv6-enabled user accesses a website across the IPv4 Internet using a typical site, “URL/domain name, such as `www.example.com`”, the resolved source IP of the website’s domain name by the DNS should appear to the user as IPv6 instead of IPv4 source IP address.

One of the ABC Corp. primary requirements is that the go-live of the IPv6 network project be within 6 weeks. Therefore, they hired a network consultant to provide a strategic approach that can help them to achieve this goal within this limited timeframe.

Network Requirements Analysis

Based on the information provided, the following are the primary design constraints:

- ABC Corp. needs a quick (transition) solution to enable IPv6.

- ABC Corp. Internet and data center services are all located at the HQ/hub site (centralized model).

- The current MPLS VPN WAN provider does not support IPv6.

Design considerations include the following:

- The network must be end-to-end IPv6 enabled.

- Provide the ability to ABC Corp. end users to access the IPv4 Internet and IPv4-only legacy applications.

- Web services accessed over the IPv4 Internet by internal IPv6 clients must appear as if they are accessed over the IPv6 Internet.

- ABC Corp. IPv6 web services must be accessible over the IPv4 Internet.

The assumption is that network nodes across the network, end-user devices, applications, and hosts within the data center support IPv6

Design Approach

To meet the primary requirements of ABC Corp., taking into account the design considerations covered earlier, ABC Corp. can consider the following phased approach.

Phase 1

Provide fast IPv6 enablement across the network (see [Figure 10-3](#)):

- Enable IPv6 on all network nodes (dual stack), starting from the DC followed by other nodes such as WAN routers.

- Enable IPv6 routing on the network nodes (DC, WAN routes hub and spokes, and Internet edge).

- Enable Stateful NAT64 at the IPv4 Internet edge gateway toward the IPv4 Internet to provide Internet access for the internal IPv6 devices.⁴

4. Framework for IPv4/IPv6 Translation, RFC 6144

- Introduce DNS64 functionality to satisfy the requirement of hiding Internet service IPv4 source addresses to appear as if sourced from an IPv6 address (by synthesizing DNS A record into AAAA record).⁵

5. DNS64: DNS Extensions for Network Address Translation from IPv6 Clients to IPv4 Servers, RFC 6147

- Enable Static NAT64 at the DC edge nodes, where the IPv4 to IPv6 static mapping can provide internal IPv6-Enabled users to access legacy IPv4-only applications.⁶

6. Stateful NAT64: Network Address and Protocol Translation from IPv6 Clients to IPv4 Servers, RFC 6146

■ Enable Static NAT64 at the IPv4 Internet gateway where the IPv6 to IPv4 static mapping can enable external users to access ABC Corp. IPv6 web-based services over the IPv4 Internet.

■ Interconnect the IPv6 network islands (spokes/remote sites) with the HQ/hub site using an IP overlay tunneling mechanism (preferably IPv6 over mGRE IP tunneling dynamic multipoint VPN [DMVPN]) over the IPv4 MPLS VPN WAN.

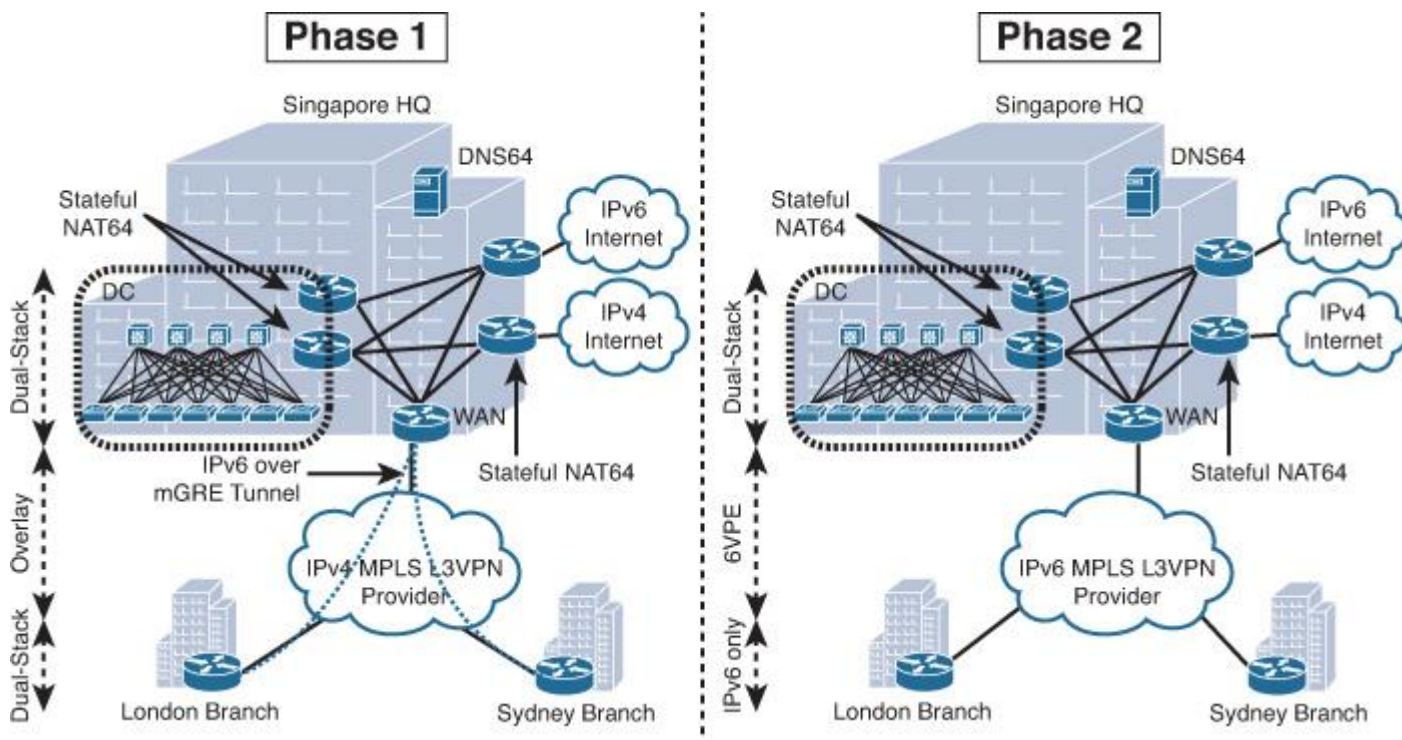


Figure 10-3 ABC Corp. Transition Approach to IPv6: Phase 1 and 2

Phase 2

Design optimization (see [Figure 10-3](#)):

■ Migrate the WAN to a provider that supports IPv6 MPLS L3VPN to be used instead of the overlaid IPv6 DMVPN over the WAN.

■ Disable IPv4 routing from the network areas where no IPv4 clients/hosts exist, such as remote sites, which helps to reduce the load from the remote site network nodes (holding separate RIB/FIB tables for each IP version).

Note

For a sample WAN migration example, see [Chapter 4, “Enterprise Edge Architecture.”](#)

Further Reading

RFC 4029, *Scenarios and Analysis for Introducing IPv6 into ISP Networks*, <http://www.ietf.org/rfc/rfc4029.txt>

“Deploying IPv6 in Broadband Access Networks,” <https://supportforums.cisco.com/docs/DOC-13985>

“Deploying IPv6 in Campus Networks (CVD),” <http://www.cisco.com>

IP Multicast Design Considerations

The typical behavior of unicast traffic is to send a copy of each packet to each receiver in any given network segment, as shown in [Figure 10-4](#). As a result, the more receivers in the network of any given application, such as streaming video, the more bandwidth consumed. The impact of this behavior in small networks is almost always acceptable because of the limited number of receivers (end users). In contrast, in large-scale enterprises and SP-grade networks, where there might be thousands or even more of receivers of certain applications such as IPTV streaming, this behavior wastes a large amount of bandwidth because of the

unnecessary extra bandwidth required to transport unicast packets, which will ultimately lead to additional cost and overutilization of the available bandwidth.

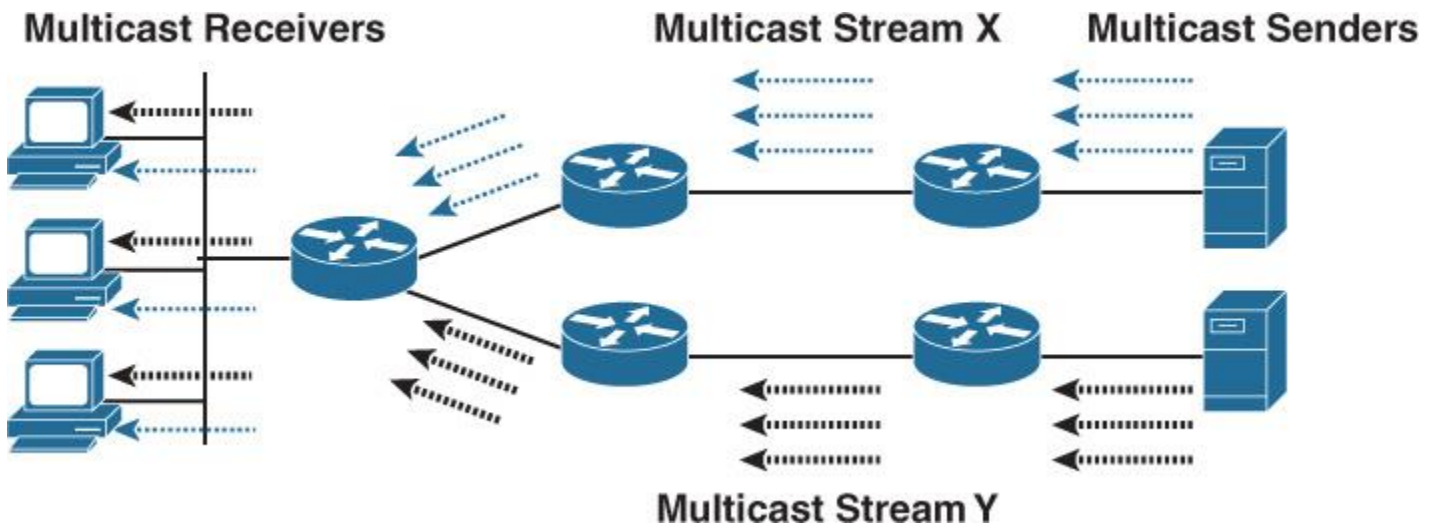


Figure 10-4 Unicast Traffic Streams

Broadcast transmissions, however, lead to even worse situations, because it will forward data packets to all portions of the network, which can consume a lot of bandwidth and hardware resources across the network when there are actually only a few intended recipients.

Therefore, in today's networks, there are many applications and services developed to operate over multicast-enabled transport to help network operators overcome the aforementioned limitations. In general, with multicast, organizations can gain several advantages, such as the following:

- **Efficient and cost-effective transport:** More efficient network and bandwidth utilization can help to reduce the cost of network resources.

- **Optimized bandwidth and efficiency:** Enhancing the overall utilization over the network can reduce congestion caused by existing applications that are inefficiently transmitting unicast to groups of recipients, thereby allowing the network to support a larger number of recipients with simultaneous access to the application in a more efficient way, as shown in [Figure 10-5](#).

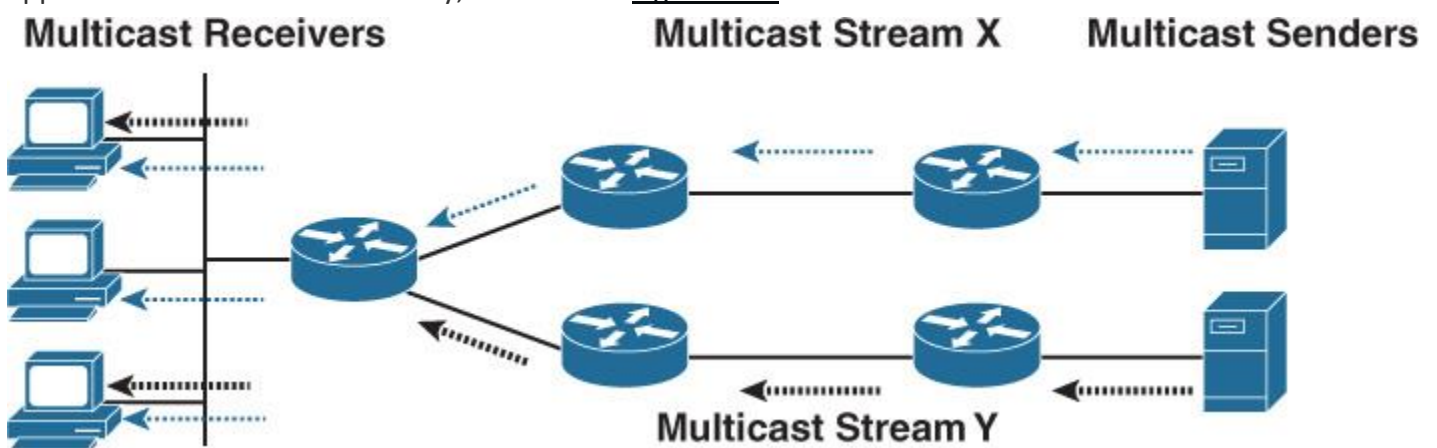


Figure 10-5 Multicast Traffic Stream

- **Provide the business access to new applications:** Many applications and services nowadays developed to operate over multicast-enabled networks that offer more efficient and scalable communication, specifically distributed multimedia business applications such as trading, market data, distance learning, and videoconferencing.

- **Open new business opportunities:** Opens new revenue-generating opportunities for both enterprises and SPs, by offering innovative services that were not possible over unicast transport, such as content SPs and IPTV.

Note

Practically and ideally, the decision to enable multicast should be driven by business application requirements. However, as a network architect or designer, you can sometimes suggest that the business migrate certain applications from the unicast version to multicast version if the option exists and the transition is going to optimize the network and application performance. For instance, Moving Picture Experts Group (MPEG) high-bandwidth video applications usually consume a large amount of network bandwidth for each stream. Therefore, enabling IP multicast will enable you to send a single stream to multiple receivers simultaneously. This can be seen from the business point of view as a more cost-effective and bandwidth-efficient solution, especially if the increased bandwidth utilization translates to increased cost (such as WAN bandwidth).

Enterprise Multicast Design Options and Considerations

This section focuses on multicast design considerations on typical enterprise networks.

Application Characteristic

Designing IP multicast is just like all other IP network designs. To be a business-driven design and facilitate business goals, it has to follow the top-down approach by first focusing on the business goals and needs, taking into account the characteristics of the applications to be used. With regard to multicast applications, they can be categorized into three primary types, as listed in [Table 10-6](#).

Application Characteristic	Application Examples
One to many	Newsfeeds, push media, paging, announcements, some database updates
Many to many	Multimedia conferencing, multiplayer games, concurrent processing
Many to one	Data collection, polling, resource discovery

Table 10-6 Multicast Applications Types

Therefore, understanding at a high level the characteristics of the multicast applications is essential to determine which multicast protocol and design might be appropriate.

Multicast IP Address Mapping into Ethernet MAC Address

Normally, in any IP multicast-enabled environment, hosts (receivers) are required to receive a single traffic stream that has a common destination MAC address that maps to a multicast channel or group. Therefore, a mapping of IP to MAC is required here per multicast destination group IP. The range from 0100.5e00.0000 through 0100.5e7f.ffff is the available range of Ethernet MAC addresses for IPv4 multicast, and only half of these MAC addresses are available for use by IP multicast, where the IP to MAC mapping will place the lower 23 bits of the IPv4 multicast group address into these available 23 bits in the MAC address. As a result, MAC addressing will not always be unique in Layer 2 switched networks, because the upper 5 bits of the IP multicast address are dropped as part of the IP to MAC mapping,⁷ as shown in [Figure 10-6](#).

⁷. "IP Multicast Technology Overview," <http://www.cisco.com>

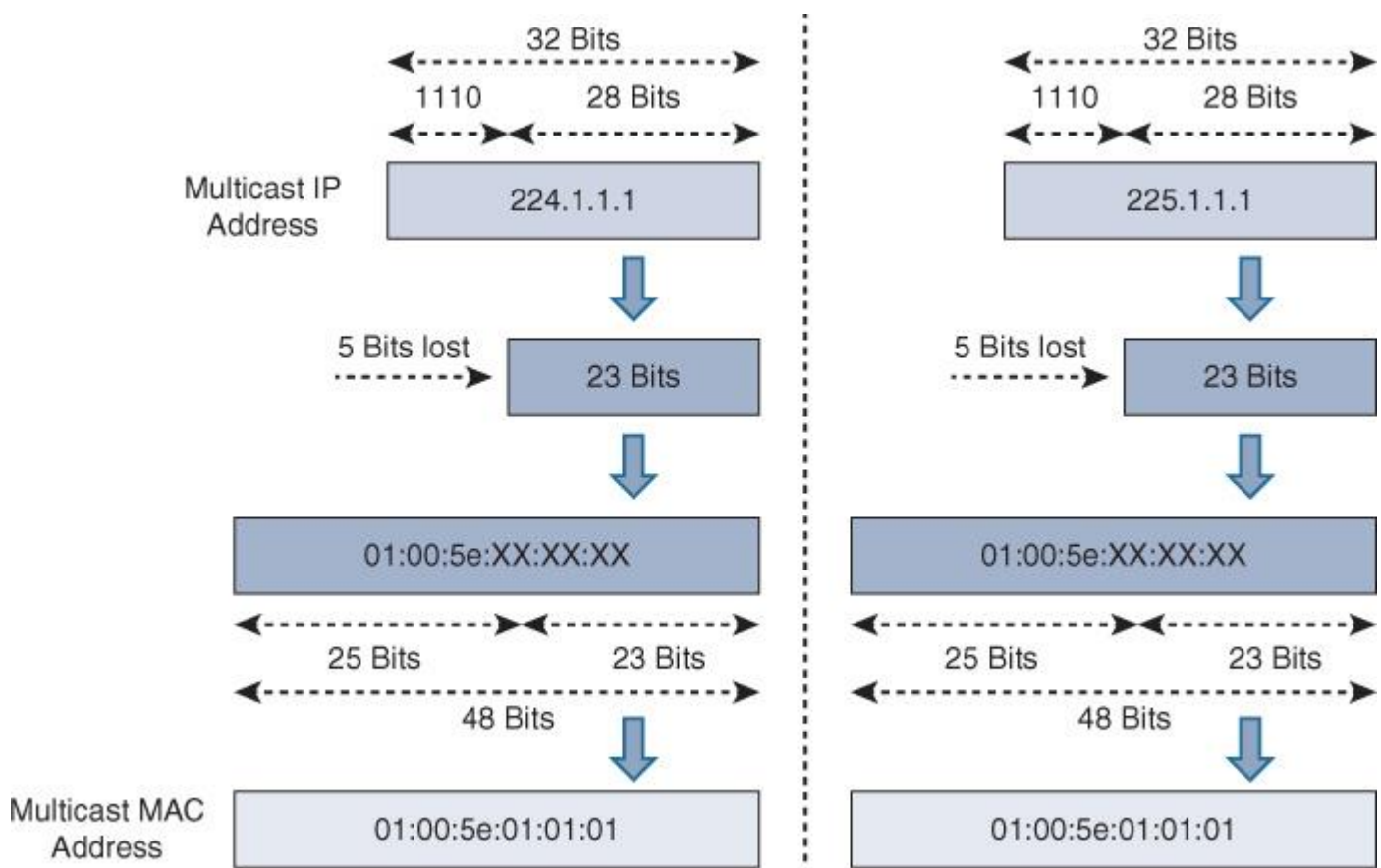


Figure 10-6 Multicast Group Layer 3 IP-to-Ethernet MAC Mapping

In other words, as shown in [Figure 10-6](#), the first 4 bits of the IP multicast 32 bits are always set to 1110 (Class D). Therefore, there will be only 28 unique bits left of the IP multicast addresses information. Furthermore, because the remaining 28 unique bits that belong to the Layer 3 IP multicast address information cannot be mapped in a 1:1 manner to the available 23 bits of Layer 2 MAC address space, this will lead to the loss of 5 bits of the address information in this mapping process. Consequently, it will result in what is known as *32:1 address mapping*, which means that each Layer 2 multicast MAC address can potentially map to 32 IP multicast addressees.

To illustrate this point, consider a scenario of two IPTV channels streaming over different IP multicast group IDs (per channel), such as 224.1.1.1 and 225.1.1.1. Based on the mapping mechanism described earlier, hosts connected to a common Layer 2 switch and interested in receiving different IPTV channels will end up receiving both streams because both multicast groups' IPs will map to the same multicast MAC address on the Layer 2 switch. In this particular example, this will usually result in reduced efficiency in a large network with a large number of IPTV receivers connected to the same Layer 2 switched network.

Similarly with IPv6, the mapping will be from 128-bit Layer 3 IPv6 multicast addresses to the 48-bit MAC address, where only the lower 32 bits of the Layer 3 address are preserved, while 80 bits of information are lost during the mapping process. This means, like IPv4, there is a possibility that more than one IPv6 multicast address might map to the same 48-bit MAC address.

Therefore, even though this might not always be an issue in LAN environments, it is still a critical point to be considered by network designers with regard to IP multicast design over a switched network. This can be even more critical if the switched network is connecting two different multicast domains (interdomain multicast scenario).

Internet Gateway Management Protocol (IGMP) snooping offers a simple solution for Layer 2 switched networks by eliminating the flooding to every single port (avoiding broadcast forwarding style). With IGMP snooping, the switch intercepts IGMP packets as they are being flooded and uses the information in the IGMP packets to determine which host/port should receive packets directed to a multicast group address.

Similarly, Multicast Listener Discovery (MLD) snooping must be considered in a Layer 2 switched network when IPv6 multicast is used.

Nevertheless, even when multicast optimization protocols in a Layer 2 network are used, such as IGMP snooping, Layer 2 switches may flood all multicast traffic that belongs to the MAC address range of 0x0100.5E00.00xx, which belongs to the respective Layer 3 addresses in the link-local block (reserved for the use of routing protocols and topology discovery protocols) to all the ports within the same Layer 2 segment on the Layer 2 switch. Therefore, when possible, IP multicast addresses that map to the MAC address range 0x0100.5E00.00xx should be avoided whenever possible.

Moreover, in scenarios such as the one shown in [Figure 10-7](#), when there are multiple Layer 3 routers connected over a Layer 2 switch using one shared segment / VLAN, IGMP snooping cannot optimize multicast traffic flooding within the layer 2 switch. This is because these routers technically do not send IGMP membership reports for the desired multicast flows. Instead, the routers use routing protocol control messages among them, such as PIM. As a result, there will be no IGMP messages for IGMP snooping to intercept and use for multicast traffic forwarding optimization, and the multicast video stream will be forwarded to all the routers connected to this switch within the same VLAN.

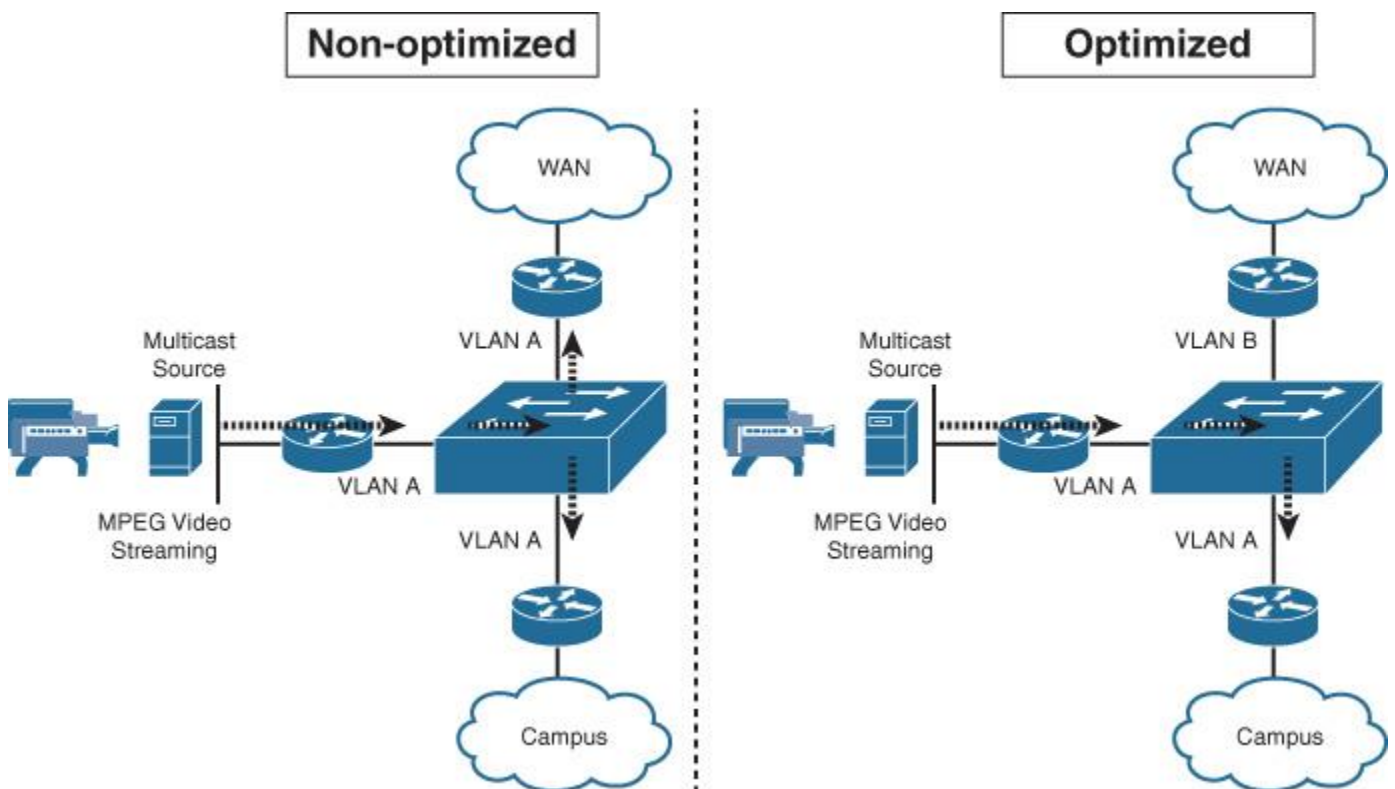


Figure 10-7 Design Issue: IGMP Snooping

To optimize this design and overcome this issue, routers that are not supposed to receive the multicast stream simply can be placed in a different VLAN in this switch, such as the router connected to the WAN in [Figure 10-7](#). Alternatively, the IGMP mroute proxy feature may be used to convert PIM join messages back into IGMP membership reports.

Note

Network designers must consider maintaining the existing Layer 3 unicast communication between the WAN router and other routers in this network after placing the WAN router in a different VLAN, such as adding Layer 3 VLAN interfaces or adding additional interfaces/subinterfaces from other routers within the same VLAN.

Note

IGMP snooping may maintain forwarding tables based on either Ethernet MAC addresses or IP addresses. Because of the MAC overlapping issues covered earlier with regard to mapping IP multicast group address to Ethernet addresses, the forwarding based on IP address is desirable if a switch supports both types of forwarding mechanisms.

Multicast Layer 3 Routing

This section starts by discussing the key consideration to achieve successful multicast pack forwarding, and then covers the most common protocols used to route multicast traffic within a single multicast domain (and also discusses multicast routing between different multicast domains).

Reverse Path Forwarding

It is critical to understand the concept of reverse path forwarding and how it can impact multicast traffic forwarding and design. Reverse path forwarding (RPF) is the mechanism used by Layer 3 nodes in the network to optimally forward multicast datagrams. The RPF algorithm uses the rules shown in the [Figure 10-8](#) to decide whether to accept a multicast packet, forward it, or drop it.

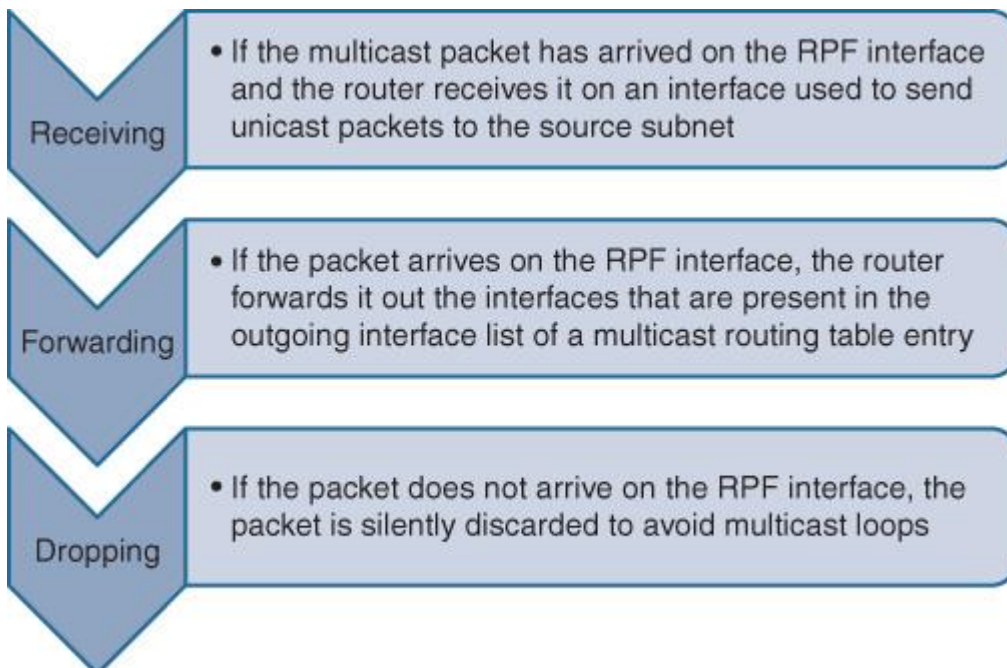


Figure 10-8 Multicast RPF Check Rules

Therefore, a successful RPF check is a fundamental requirement to establishing multicast forwarding trees and passing multicast content successfully from sources to receivers. For instance, in the scenario shown in [Figure 10-9](#), multicast was enabled only on interface G0/1, while unicast routing prefers interface G0/0 to reach the multicast sender (host IP 192.168.1.10). Hence, multicast traffic will not be received in this case because of RPF check failure. This issue can be fixed in different ways, such as aligning multicast-enabled interfaces with the unicast routing table, using static multicast route (mroute) or MP-BGP to overcome the issue associated with the failure of RPF check. However, the primary goal here is first to be aware of the RPF impact. Then, as a network designer, you can decide which mechanism is more feasible to use to avoid the failure of RPF check. In other words, to achieve a successful IP multicast design, RPF check must be considered along with the possible mechanism that helps to avoid any forwarding failure while at the same time minimizing increased control plane complexity. For example, if BGP is used as the unicast control plane protocol, MP-BGP will avoid adding extra control plane complexity, because one routing protocol will be

used to control both unicast and multicast routing in this case. Technically, MP-BGP here will only help to avoid RPF check failure over the desired interface to be used for multicast forwarding when the existing unicast routing prefers another path or interface.

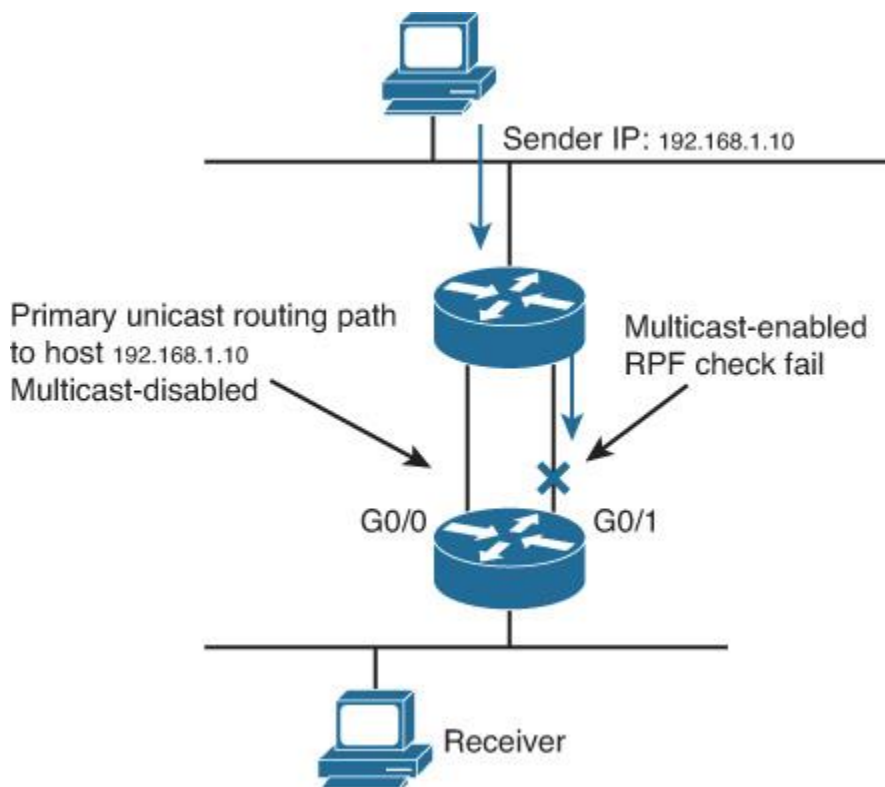


Figure 10-9 RPF Check Failure Scenario

Multicast Routing Protocols

Multicast traffic flows are IP traffic, which almost always traverses the network using the existing unicast routing. However, for the multicast receivers and senders to locate each other across the network and communicate using the underlay IP network, a multicast routing protocol must use this function, and usually each multicast routing protocol has its own characteristics in terms of the discovery and forwarding style (multicast trees) of IP multicast traffic streams. [Table 10-7](#) compares the most common multicast routing protocols.

Multicast Protocol Type	IP Version	Supported Multicast Trees	Rendezvous Point (RP)	Suitable Applications	Supported IGMP/MLD Version	Scope
PIM-DM	4	Source tree only	Not required	Legacy	IGMP 1, 2 and 3	Intradomain
PIM-SM	4/6	Shared tree and source tree	Required	One to many	IGMP 1, 2, and 3 / MLD 1, 2	Intradomain and interdomain
PIM-SSM	4/6	Forwarding only use shortest path first (SPT)	Not required	One to many	IGMP 3 / MLDv2	Intradomain and interdomain
PIM-Bidir	4/6	Shared tree only	Required	Many to many Many to one	IGMP 1, 2, and 3 / MLD 1, 2	Intradomain
PIM-BSR (PIMv2)	4/6	Shared tree and source tree	Required	One to many	IGMP 1, 2, and 3 / MLD 1, 2	Intradomain and interdomain

Table 10-7 Comparison of Layer 3 Multicast Routing Protocols

Note

Although some of the PIM protocols covered in [Table 10-7](#) technically support multicast interdomain routing, it is not common for it to be used to provide multicast interdomain routing without other protocols such as Multicast Source Discovery Protocol (MSDP).

Although having multiple flavors of PIM might be seen like an added complexity to the multicast network design, it can be seen as added flexibility for an environment with different types of multicast applications. For instance, PIM-SSM can be deployed for certain enterprise communication applications, PIM-Bidir for financial applications, and PIM-SM for other general IP multicast communications.

RP discovery, however, is one of the primary design aspects that must be considered during the planning and design phase of any IP multicast design task. [Table 10-8](#) summarizes the common mechanisms used to locate or discover the intended multicast RP within a multicast domain.

	Manual/Static	Automatic (Auto-RP)	BSR	Embedded RP
Supported IP version	Support IPv4/v6.	IPv4	IPv4/IPv6	IPv6
Scalability	Scalable, with a potential of increased operational overhead.*	Scalable	Scalable	Scalable
Operational simplicity and flexibility	Inflexible with management overhead in large deployments. Changing RP's IP requires all the nodes to be updated.	Flexible with simplified operation.** Offers more protocol's filtering capabilities than BSR	Flexible with simplified operation. Offers less traffic overhead (as RP information is encapsulated in PIM packets)	Flexible with simplified operation, dynamic

*If static RP used with a redundant RP mechanism such as anycast-RP with MSDP or anycast-RP PIM (RFC 4610), static RP can offer a scalable solution without significant increase in the operational overhead.

** Taking into account traffic and processing overhead because of the specific multicast groups used to propagate RP information, also special consideration in hub-and-spoke topology "NBMA" is required.

Table 10-8 Multicast RP Discovery Mechanisms

Note

Auto-RP is a Cisco proprietary protocol. Bootstrap Router (BSR), in contrast, is the standards body method of electing an RP (RFC 5059). BSR operates completely based on PIMv2 and PIM-SM, which offer an optimized bandwidth during the flooding and discovery process as compared to Auto-RP. In contrast, Auto-RP can scope multiple RP address per domain and operate over either PIMv1 or PIMv2, with the ability to fall back to dense mode if required. As a general rule, network designers should avoid considering BSR and Auto-RP protocols within the same domain at the same time. That said, in some scenarios where a multicast PIMv2 domain that does not support Auto-RP (for example, non-Cisco network nodes) needs to interoperate with a multicast PIMv1 domain running Auto-RP, both Auto-RP and BSR functions are required. One simple solution to facilitate the interoperability between these two different multicast PIM domains is to deploy a network node at the multicast domain's boundary to perform the Auto-RP mapping agent and the BSR functions. Also, this approach is applicable to PIM migration scenarios between Version 1 and Version 2.⁸

8. http://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst3750x_3560x/software/release/15-2_2_e/multicast/configuration_guide/b_mc_1522e_3750x_3560x_cg/b_mc_3750x_3560x_chapter_010.html

Note

The RP is required to initiate new multicast sessions with sources and receivers. During these sessions, the RP and the first-hop router (FHR) may experience some periodic increased overhead from processing. However, this overhead will vary based on the multicast protocol in use. For instance, the RP with PIM-SM Version 2 requires less processing than in PIM-SM Version 1, because the sources only periodically register with the RP to create state. In contrast, the

location of the RP is more critical in network designs that rely on a shared tree, where the RP must be in the forwarding path, such as PIM-Bidir. (The following section covers this point in more detail.)

RP Placement

Normally, the placement of the a multicast RP is influenced primarily by the following factors:

- The used multicast protocol
- The multicast tree model
- Application characteristics (for example, one to many versus many to many)
- Targeted network between the sources and receivers (LAN versus WAN)

As a rule of thumb, when the source tree along with shortest path tree (SPT) are considered, RP placement is not a big concern even though it is commonly recommended to be placed closer to the multicast sources, because in this case the RP is not necessarily required to be in the data forwarding path. However, some multicast applications, such as many-to-many type of multicast applications the recoveries, might operate as senders at the same time using different multicast groups for receiving and sending. Technically, even when SPT is enabled (where the last-hop router [LHR] cuts over to SPT source tree), the source tree is always created between the source and the RP, in which a (S,G) state is created on all the nodes between each source and its RP before the switch over to SPT takes place, as shown in [Figure 10-10](#). This may lead the nodes in the path (between the many receivers/senders and the RP) to hold a large number of (S,G) states. (In trading environments, this number can reach up to a few thousands of source feedback streams sourced from the receivers that operate as multicast senders as well.)

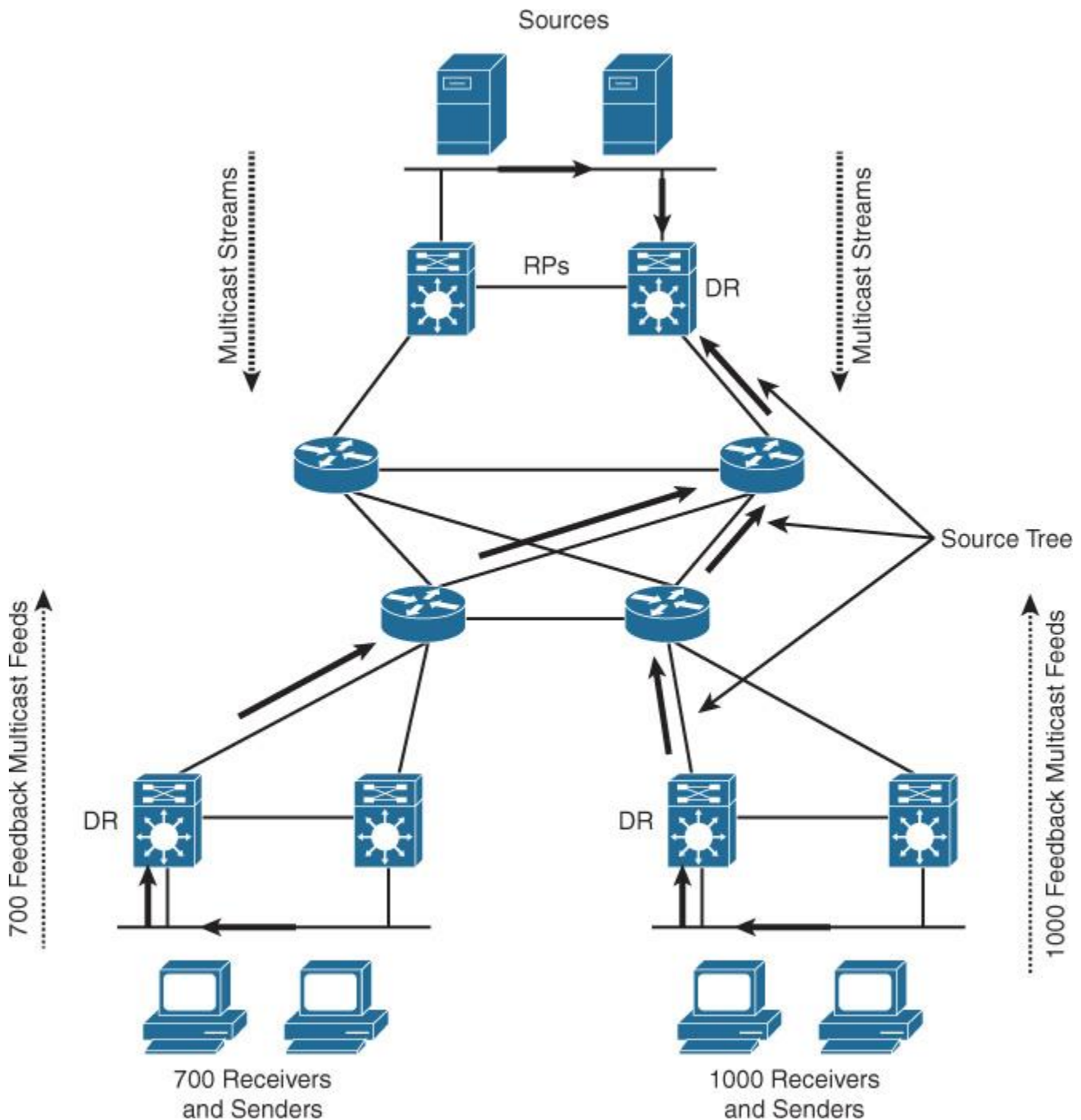


Figure 10-10 Many-to-Many Multicast Applications Using PIM-SM RP

Therefore, to reduce the number of (S,G) states on these nodes, you can add different RPs close to the receivers that require sending feeds to the feedback groups, as shown in [Figure 10-11](#). Also, MSDP can be introduced in this design among the RPs to ensure that all RPs for any given group will be aware of other active sources. This design is suitable in a multicast environment that requires the receivers to be able to send feedback/streams using a separate group from the actual data source group. Alternatively, PIM-SM can be migrated to PIM-Bidir in this environment.

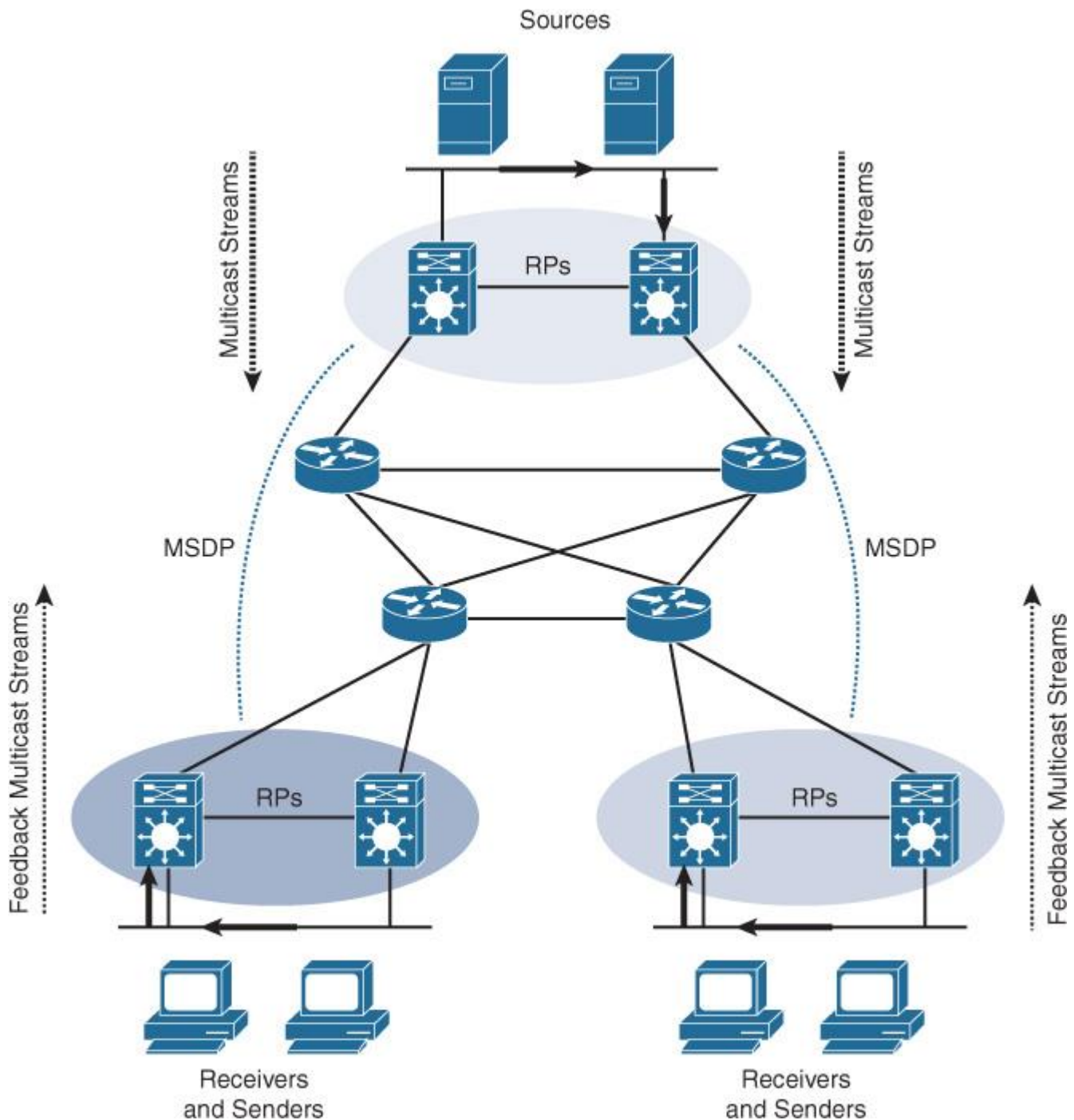


Figure 10-11 *Optimized - Many-to-Many Multicast Applications Using PIM-SM RPs*

However, when the shared tree is used as the forwarding multicast tree, such as using PIM-Bidir where the RP will be in the data path, network designers must carefully and wisely consider the placement of the RP within the network because all the traffic will flow through the RP. For example, if multiple multicast streams are sourced from different senders distributed at different locations across the network, identifying which stream should be given priority is a key to place the RP in the most optimal path between the sources and receivers based on the following:

- Application's criticality to the business
- Bandwidth consumption of each application's stream
- Number of sources and receivers of each application
- The underlying transport and topology of the network between the sources and receivers (LAN versus WAN, hierarchical versus hub and spoke)

For example, in the scenario shown in [Figure 10-12](#), there are two hub nodes each connected to a different data center with multicast applications streaming over a shared tree. Although hub 1 is connected directly to the business-critical multicast application that requires high bandwidth, multicast streams will reach each of the remote site's receivers via hub 2. This is because hub 2 is defined as the RP for the multicast shared tree (such as PIM-Bidir), where the RP must be in the data forwarding path, which can result in a congested data center interconnect (DCI) link and degraded application quality. Therefore, to optimize the path for the business-critical application with high-bandwidth requirements, the RP function must be moved to hub 1 to serve as an RP either for the entire multicast domain or at least for the multicast groups used by the applications located in DC-1. This can provide a more optimal path for multicast streams sourced from DC-1 toward the spokes/receivers.

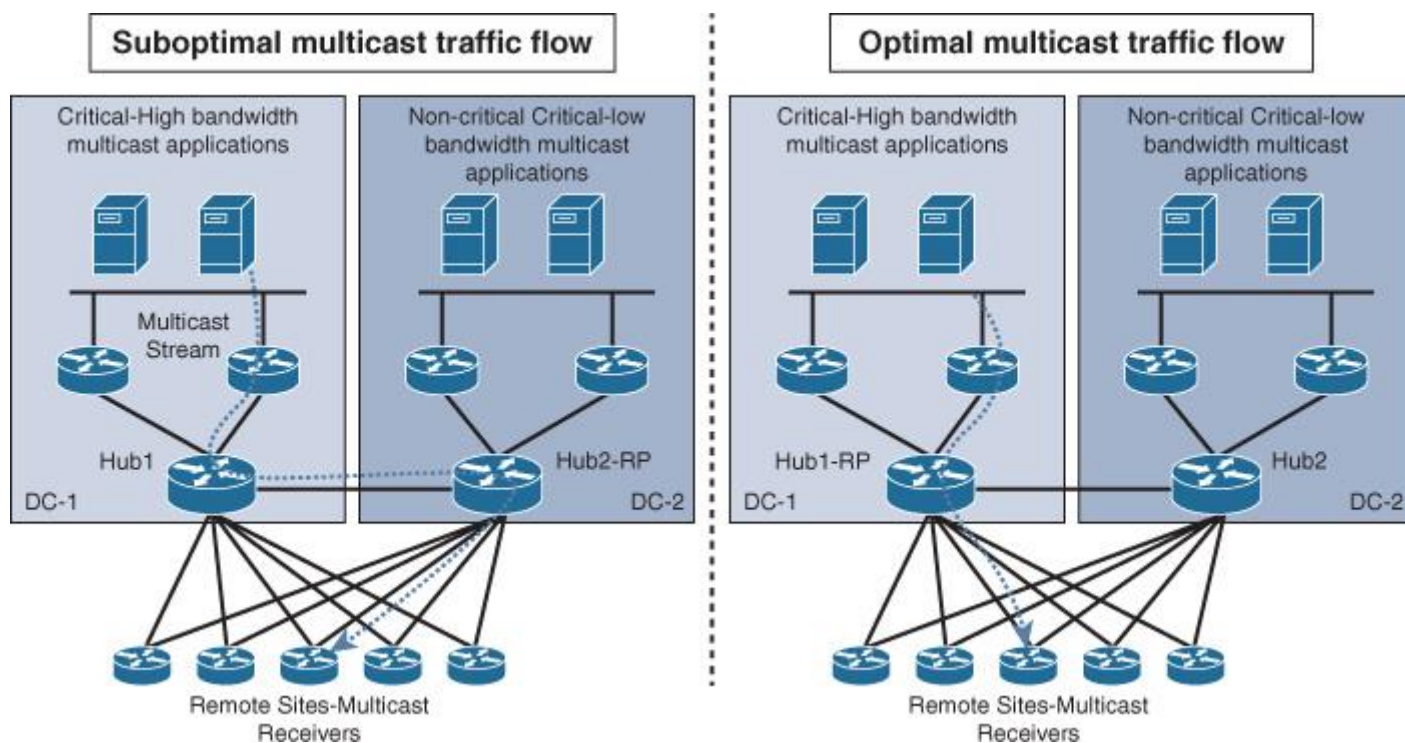


Figure 10-12 RP Placement Consideration: Shared-Tree Multicast

Note

In both scenarios, the assumption is that the RPF check is considered based on the utilized path for multicast traffic.

Interconnecting Multicast Islands

One of the most common multicast design challenges occurs when two or more multicast-enabled networks need to communicate over a unicast-only network—for example, when a retail business is planning to roll out a digital media signage solution across all the remote sites, where the media server of this solution will stream the media contents centrally from the data center to the remote sites “acting as a multicast streaming source.” This digital media signage solution requires multicast to be enabled end to end. However, the WAN provider in this scenario supports only unicast. In this case, this enterprise either needs to migrate to a new WAN provider or transport that supports multicast or consider a tunneling mechanism such as GRE in which IP multicast traffic can ride this tunnel and be transported over the IP unicast WAN network, as shown in [Figure 10-13](#). Moreover, if the network in the middle is MPLS enabled, this may facilitate building a self-deployed L2VPN (such as point-to-point E-Line), then on top of it multicast routing can be enabled and transported.

Multicast over GRE/LTPv3/DMVPN

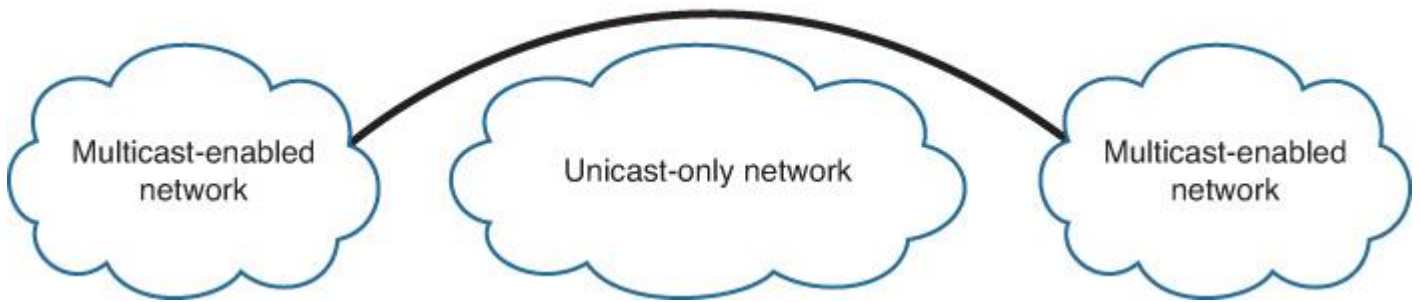


Figure 10-13 *Connecting Multicast Islands*

Which decision is better depends entirely on the situation and the requirements, such as timeframe to deploy the solution, cost considerations, and equipment support to the overlay technology. (For example, do the WAN edge nodes support GRE?) Furthermore, the most important influencing factor here is the characteristics of the multicast application used. For instance, IP tunneling such as GRE or DMVPN may introduce some limitations, such as limited scalability if a many-to-many multicast application is used. Therefore, in this case, it might be more feasible for large-scale organizations to consider a core/WAN transport that supports multicast routing rather than considering a solution that may potentially introduce limitations to the business in the future.

The other common scenario here is when there is a network device in the network path that does not support multicast, such as a firewall. This device will usually break the multicast network into two networks, where a mechanism such as IP tunneling is required to pass multicast traffic through this device, as shown in [Figure 10-14](#), assuming this approach (tunneling traffic through the firewall) will not breach the organization's security policy standards.

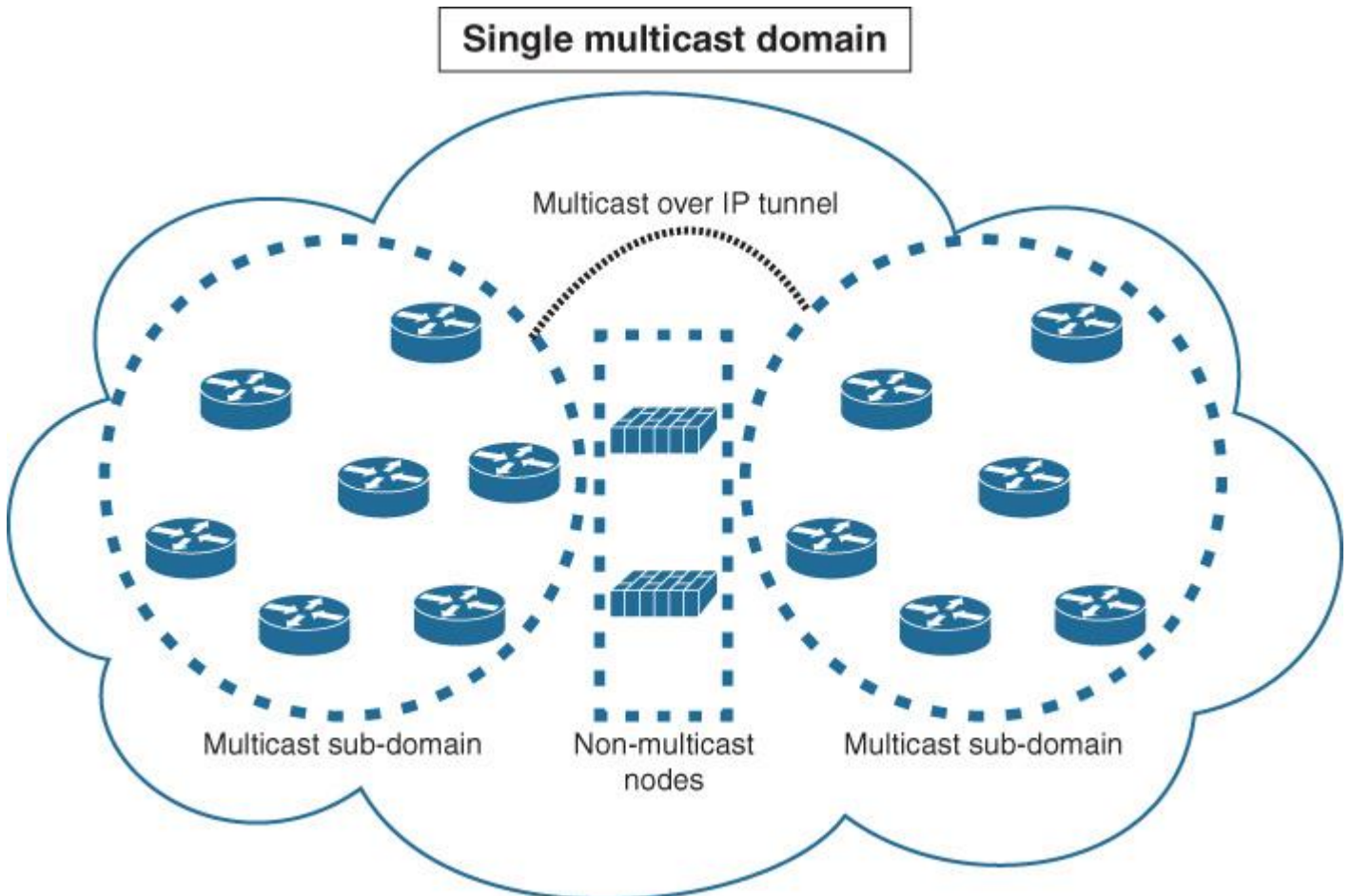


Figure 10-14 *Connecting Multicast Islands over a Non-Multicast-Enabled Node*

Interdomain Multicast

The multicast protocols discussed earlier focused mostly on handling multicast in one multicast domain. The term *multicast domain* can be defined as an interior gateway protocol (IGP) domain, one BGP autonomous system (AS) domain, or it can be based on the administrative domain of a given organization. For example, one organization might have multiple multicast domains, with each managed by a different department; for instance, one domain belongs to marketing, and another domain belongs to engineering. Other common scenarios of multiple multicast domains are between service providers and after merger or acquisitions between companies. Therefore, it is important sometimes to maintain the isolation between the different multicast domains by not sharing streams and RP feeds and at the same time offering the ability to share certain multicast feeds and RP information as required (in a controlled manner). This section covers the most common protocols that help to achieve this type of multicast connectivity.

Multicast BGP

As discussed earlier, a successful RPF check is a fundamental requirement to establish multicast forwarding trees and pass multicast content successfully from sources to receivers. However, in certain situations, unicast might be required to use one link and multicast to use another, for some reason, such as bandwidth constraints on certain links. This situation is common in interdomain multicast scenarios. Multicast Border Gateway Protocol (MP-BGP and sometimes referred to as MBGP as well) is based on RFC 2283, “Multiprotocol Extensions for BGP-4”. MP-BGP offers the ability to carry two sets of routes or network layer reachability information (NLRI) (sub-AFI) set for unicast routing and one set (NLRI) for multicast routing. BGP multicast routes are used by the multicast routing protocols to build data distribution trees and influence the RPF check. Consequently, service providers and enterprise networks can control which path multicast can use and which path unicast can use using one control plane protocol (BGP) with the same path selection attributes and rules (such as AS-PATH, local_preferance, and so on).

Multicast Source Discovery Protocol

MSDP is most commonly used to provide a mechanism of RP redundancy along with Anycast-RP (covered later in this chapter). MSDP is the most common protocol used to interconnect multiple IPv4 PIM domains because of its controlled and simplified approach of interconnecting PIM domains, in which it allows PIM domains to use an interdomain source tree instead of a common shared tree. With PIM, each RP is usually aware of the multicast sources and receivers within its PIM boundary (domain). MSDP peers use TCP sessions and they send Source Active (SA) messages to inform other MSDP peers about an active source within the local multicast domain, as shown in [Figure 10-15](#).

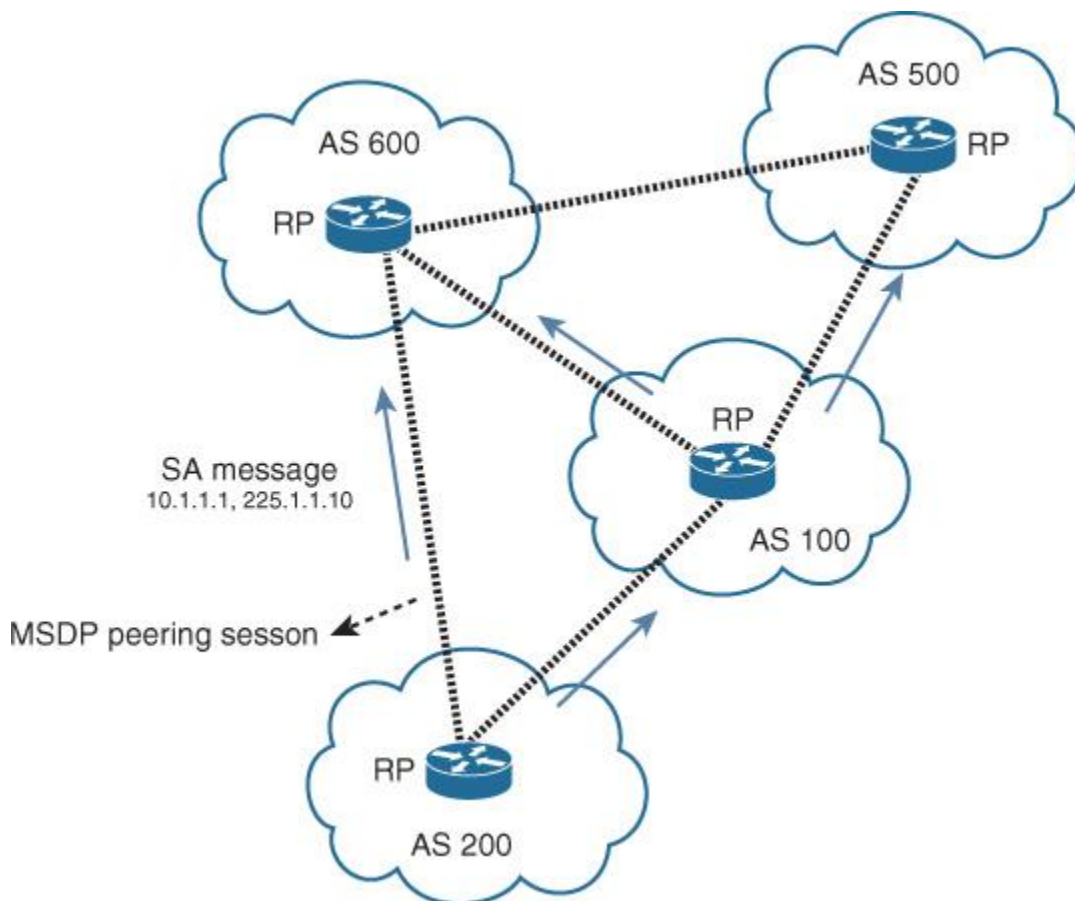


Figure 10-15 Typical Interdomain Multicast Scenario with MSDP

Consequently, all the MSDP peers (RPs) will be aware of all sources within the local domain and in other domains. RPF check is a fundamental consideration here, as well. However, with MSDP, some rules drive whether RPF check for SA messages is to be performed, as summarized in [Table 10-9](#) [106].

RPF Check Performed	RPF Check Not Performed
The sending MSDP peer is also an interior (MP-)BGP peer.	The sending MSDP peer is the only MSDP peer (for example, if only a single MSDP peer or a default MSDP peer is configured).
The sending MSDP peer is also an exterior (MP-)BGP peer.	The sending MSDP peer is a member of a mesh group.
The sending MSDP peer is not an (MP-)BGP peer.	The sending MSDP peer address is the RP address contained in the SA message.

Table 10-9 MSDP RPF Check Rules

One common design issues with regard to multidomain multicast using MSDP and BGP that leads to RPF failure is that the IP address of the i(MP-)BGP peer is different from the MSDP IP (for example, BGP using physical IP, while MSDP using loopback IP), as shown in [Figure 10-16](#).

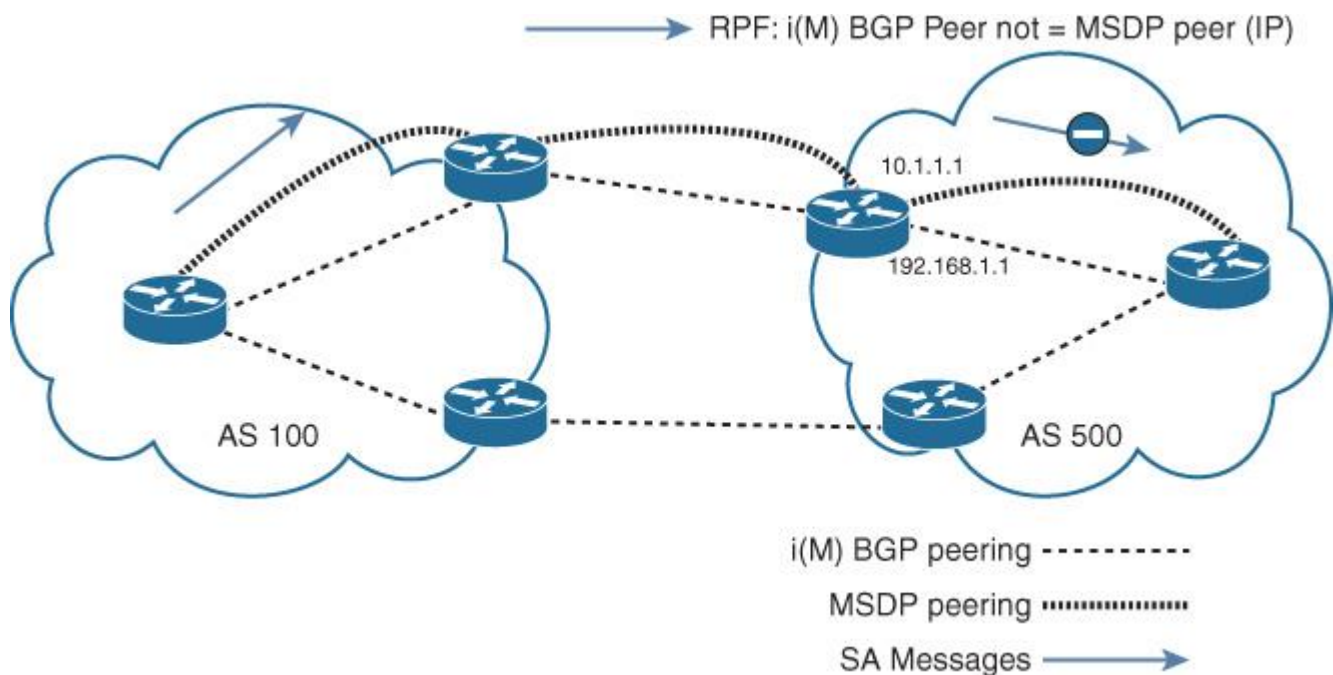


Figure 10-16 Common RPF Check Failure with MSDP and BGP

Therefore, it is important that the address used for both i(MP-)BGP and MSDP peer addresses is the same.

Interdomain Multicast Design Scenario

In the scenario shown in [Figure 10-17](#), AS 500 is providing transit connectivity service to a content provider (AS 300) that offers IPTV streaming. End users need to connect to the streaming server IP 10.1.1.1 over AS 500. AS 500 has two inter-AS links, and they want to offer this transit service with AS 300 using the following traffic engineering requirements:

- Unicast traffic between AS 500 and AS 300 must use the link with 5G as the primary path and fail over to the 10G link in case of a failure.
- Multicast traffic must use the 10G link between AS 500 and AS 300. However, in case of a link or node failure, multicast traffic must not fail over to the other link (to avoid impacting the quality of other unicast traffic flowing over the 5G inter-AS link).
- Multicast group addresses that are in the range of 232/5 must not be shared between the two domains (AS 300 and AS 500).
- Currently, the IPTV system needs to use only the range of 225.1.1.0/24. Therefore, only sources in AS 300 with this range must be accepted by AS 500.

To achieve these requirements, the following design aspects must be considered:

- To ensure multicast traffic flow is over the 10G link only and without facing RPF check failure, MP-BGP will be used to advertise the multicast sources IPs (e.g. 10.1.1.1) and filter out these IPs from being advertised/received over the 5G link.
- MSDP peering must be established between multicast RPs of AS 300 and AS 500 to exchange SA messages about the active source within the local domain in each AS.
- PIM RP filtering and MSDP filtering⁹ is required to ensure that the RPs will only send/accept sources within the multicast group IP range (225.1.1.0/24).

⁹. "Multicast Source Discovery Protocol SA Filter Recommendations," <http://www.cisco.com>

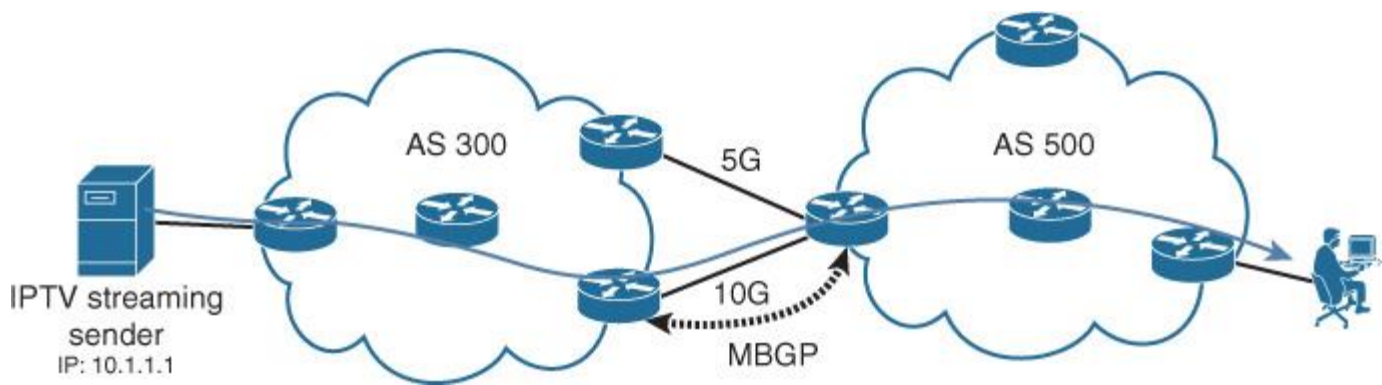


Figure 10-17 Interdomain Multicast Design Scenario

Embedded RP

Although PIM SSM offers the ability for IPv6 multicast to communicate over different multicast domains, PIM SSM still does not offer an efficient solution for some multicast deployments where few-to-many and many-to-many types of applications exist, such as videoconferencing and multi-user games applications. Also, in some scenarios, the multicast sources between domains may need to be discovered. Furthermore, MSDP cannot be used to facilitate interdomain multicast communication as with IPv4, because it has deliberately not been specified for IPv6. Therefore, the most common and proven solution (at the time of this writing) to facilitate interdomain IPv6 communication is the IPv6 Embedded-RP (described in RFC 3306) “in which the address of the RP is encoded in the IPv6 multicast group address, and specifies a PIM-SM group-to-RP mapping to use the encoding, leveraging, and extending unicast-prefix-based addressing.”¹⁰ The IPv6 Embedded-RP technique offers network designers a simple solution to facilitate interdomain and intradomain communication for IPv6 *Any-Source Multicast* ASM applications without MSDP.

10. RFC 3306, *Embedding the Rendezvous Point (RP) Address in an IPv6 Multicast Address*

However, network designers must consider that following an RP failure event, multicast state will be lost from the RP after the failover process because there is no means of synchronizing states between the RPs (unless it is synchronized via an out-of-band method, which is not common). In addition, with MSDP, network operators have more flexibility to filter based on multicast sources and groups between the domains. In contrast, with Embedded-RP, there is less flexibility with regard to protocol filtering capabilities, and if there is no other filtering mechanism, such as infrastructure access lists, to limit and control the use of the RP within the environment, a rogue RP can be introduced to host multicast groups, which may lead to a serious service outage or information security risk.

SP Multicast Design Options and Considerations

This section focuses on the design considerations and options that apply to SP-grade networks (whether it is an enterprise or an SP).

MVPN (Draft-Rosen Model)

Traditionally, a separate function was required to support IP multicast over an MPLS VPN network because MPLS had no native ability to support it. The most commonly deployed MPLS VPN multicast solution is the MVPN based on IETF Draft-Rosen, where multiple domain trees (MDTs) are created per VPN over GRE tunneling across the MPLS network. This architecture usually uses the following concepts and components:

- Default MDT (per virtual router and forwarder (VRF)/customer).
- VPNv4 or VPNv6 needed to build up unicast control plane (to ensure that the customer multicast source IPs will be reachable and to avoid RPF check failure).
- Optionally, the DATA MDT can be created for high rate sources and to offload the PEs that do not need to join the multicast stream, as shown in [Figure 10-18](#).

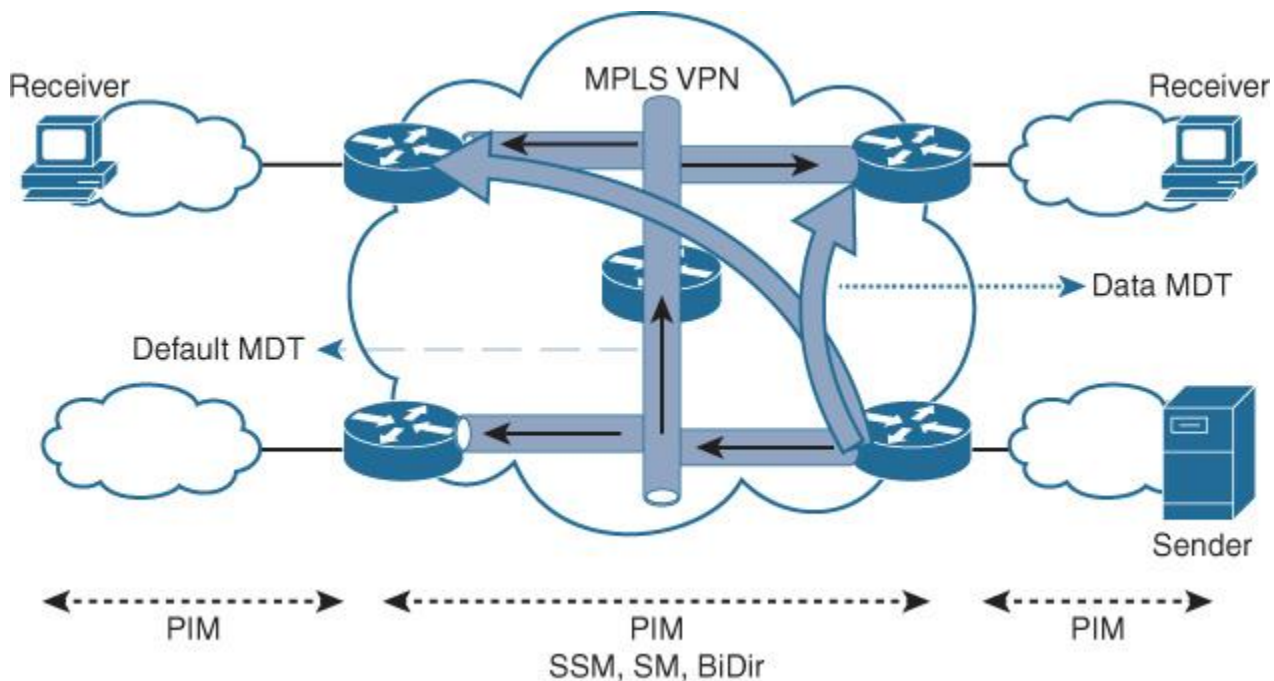


Figure 10-18 MVPN Draft-Rosen Based

In fact, this approach is the most widely deployed approach, not because it is the best approach, but because of the limited options. With this approach, for example, all PEs will receive multicast traffic on the default MDT that is attached to a given VPN, even if the PE has no interested receiver in that VRF for the (*,G) or (S,G). Therefore, the data MDT is used to overcome this issue. Furthermore, to work, this approach requires PIM to be enabled across the core. Therefore, it will increase control plane complexity of the SP core infrastructure to maintain multicast states and PIM adjacencies. In addition, each PE across the network requires PIM adjacency with each remote PE (per VPN/VRF). Therefore, new approaches were introduced to offer more scalable and flexible design options to meet next-generation (NGN) requirements. These options are discussed in the subsequent sections.

Note

This model is the most widely deployed MVPN model at the time of this writing. This means that it is a proven model with a larger number of people who have expertise to design, deploy, and operate it. Therefore, regardless of the limitations mentioned previously, many SPs and enterprises (with SP network style) still prefer to deploy the MVPN (Draft-Rosen) model because of staff expertise and knowledge.

MVPN - Label Switch Multicast

Label Switch Multicast (LSM), defined in RFC 5332, offers the next generation of IP multicast over MPLS networks, where packets are transported using MPLS encapsulation. With LSM, service providers now have the flexibility to run both unicast and multicast together while sharing the same label space. In addition, the two primary MPLS protocols, Resource Reservation Protocol Traffic Engineering (RSVP-TE) and Label Distribution Protocol (LDP), were modified to support point-to-multipoint (P2MP) and multipoint-to-multipoint (MP2MP) label-switched paths (LSPs), respectively. The MVPN LSM-based model brings several benefits, such as the following:

- LSM simplifies overall control plane design complexity through its ability to share the same control plane of the unicast routing.
- One unified forwarding plane for both unicast and multicast using MPLS encapsulation.
- Offers the ability to use advanced unicast MPLS features such as fast reroute (FRR) and bandwidth reservation.

Foundation Protocols to Build Multicast LSPs

MLDP and RSVP-TE P2MP are the primary protocols that can build multicast LSPs in MPLS networks. Although both protocols are used for the same general purpose with regard to LSM, each has its own characteristics, as listed in [Table 10-10](#).

MLDP	RSVP-TE P2MP
LSPs are built from the leaf to the root.	LSPs are built from the headend to the tail end, which may require manual static joins on the tunnel.
Supports P2MP and MP2MP LSPs (trees).	Supports only P2MP LSPs (tree).
FRR requires setup of RSVP-TE unicast backup path.	Native support of traffic engineering (bandwidth reservation, FRR).
No periodic signaling because it is reliable over TCP (hard state).	Signaling is periodic (soft state)
Control plane signaling over either P2MP or MP2MP.	Control plane signaling is built over P2P (scalability limitations).
Forwarding plane is based on P2MP and MP2MP as well.	Forwarding plane is based on P2MP (packet replication in the core).
Can be suitable for all multicast applications, especially for many to many.	Can be suitable for video delivery. Best for one/few to many

Table 10-10 *MLDP Versus MPLS-TE P2MP*

Next-Generation MVPN

The next-generation MVPN solution introduces BGP as a control plane protocol for MVPN. After the success of unicast BGP in SP networks, SPs now have the flexibility to leverage the same unicast control plane used in their network to support multicast routing as well (BGP-based), along with the same MPLS switching using LSM to encapsulate multicast within the core in MPLS labels. BGP offers the ability to separate functions of the control plane and data plane by relying on BGP to perform PEs' automatic discovery and distributing customer multicast (C-multicast) routes between the PEs. By using one flexible control plane for both unicast and multicast (including IPv4 and IPv6), SPs can achieve the ultimate level of architecture flexibility and simplified manageability, because no new skills or expertise are required to manage two different protocols and overlaid architectures. The following are some of the primary drivers toward NG MVPN:

- Unified signaling protocol for unicast and multicast VPN that offers SPs simplified control plane architectures.
- Unified forwarding plane using MPLS labels (LSM)
- Automated learning of participating nodes (PEs) with unidirectional P-Tree
- Customers' PIM-SM carried per VRF without the need of a shared-tree creation over the SP core (PIM free core)
- Flexibility to support different trees, such as P2MP and MP2MP, which can meet different application requirements
- Supports MPLS VPN inter-AS models
- Inherits the well-known BGP stability and reliability

Figure 10-19, an MPLS L3VPN SP, is offering multicast service to its clients, in addition to content services (IPTV). By using MVPN, this provider will use the same BGP control plane architecture (for example, RR) to perform the PE discovery for interested multicast receivers and to propagate the IP C-multicast between the PEs. Typically, BGP carries the C-multicast from the receiver sites to the sender sites (among PEs), which will result in the dynamic discovery of the multicast receivers. In addition, the core is kept PIM free by using LSM across the MPLS network, where each PE can perform PIM to BGP to LSP mapping.

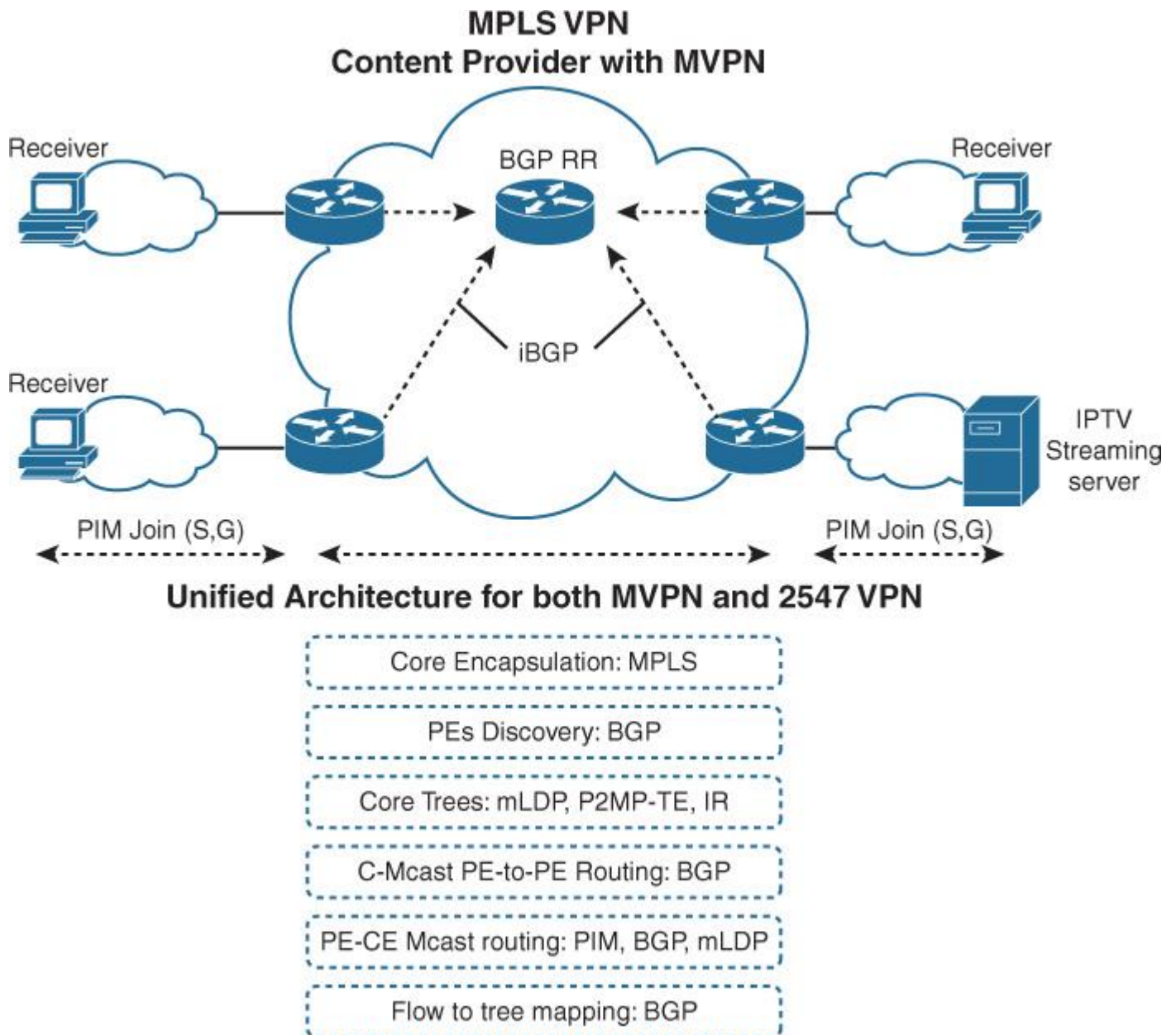


Figure 10-19 NG-MVPN High-Level Architecture

However, as covered earlier, because of the lack of expertise with this model (at the time of this writing), this model is not always the preferred option for enterprises and SPs. In addition, the support of mLDP capability is mostly available in high-end network products (SP grade). Therefore, it is not always a cheap option for enterprises to consider.

Multicast Resiliency Design Considerations

In today's modern and converged networks, it is common that several business-critical applications run over multicast, where packet or connectivity loss is a highly undesirable experience, especially if the application is serving some core business applications, such as in financial services and trading applications, where minutes of disconnectivity can translate to the loss of hundreds of thousands of dollars.

Similarly, for content providers such as the IPTV provider, the quality of the multimedia and its availability is critical to maintaining its reputation and meeting the service level agreement (SLA) requirements with the subscribers. Therefore, the design of IP multicast must incorporate the consideration of high availability and resiliency of the solution. However, with IP multicast, two levels can be considered to offer a reliable service. The first level is the underlying unicast IP infrastructure (covered in [Chapter 9, “Network High-Availability Design”](#)). The second level is the core component of the multicast network, such as RPs and other network-based solutions that are multicast specific, such as redundant sources, multicast with path diversity, and FRR. Specifically, the availability of RP in a PIM multicast domain is critical, especially for PIM-Bidir and PIM-SM. Therefore, a design for a reliable multicast-enabled infrastructure must consider the availability of the RP as well. This section covers the common mechanisms and approaches used to address multicast resiliency design requirements.

Anycast RP

The concept of Anycast RP is based on using two or more RPs configured with the same IP address on their loopback interfaces. Typically, the Anycast RP loopback address is configured as a host IP address (32-bit mask). From the downstream router’s point of view, the Anycast IP will be reachable via the unicast IGP routing. Because it is the same IP, IGP normally will select the topologically closest RP (Anycast IP) for each source and receiver. MSDP peering and information exchange is also required between the Anycast RPs in this design, because it is common for some sources to register with one RP and receivers to join a different RP, as shown in [Figure 10-20](#).

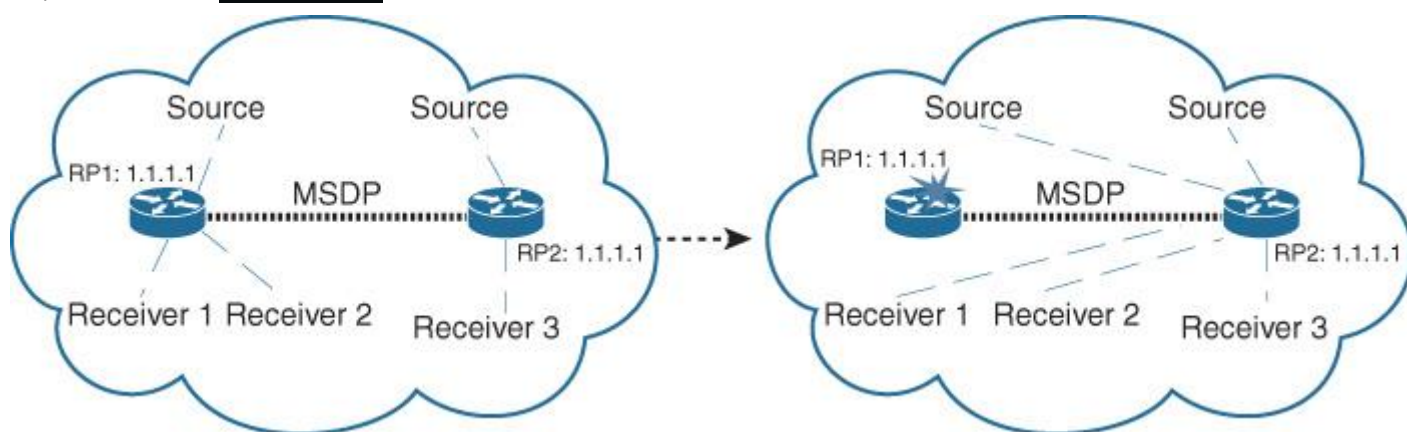


Figure 10-20 Anycast-RP with MSDP

In the event of any Anycast RP failure, IGP will converge, and one of the other Anycast RPs will become the active RP, and sources and receivers will fail over automatically to the new active RP, thereby maintaining connectivity.

Note

IPv6 does not support MSDP. Therefore, each RP has to define other RPs in the network as PIM RP-set to maintain a full list of sources and receivers in the network. Alternatively, Anycast-RP using PIM, described in IETF RFC 4610, can be used instead where the “Anycast-RP functionality can be retained without using MSDP.”

Anycast-RP Using PIM

Anycast-RP with PIM (described in RFC 4610) enables both IPv4 and IPv6 to retain the same ultimate goal network designers aim to achieve using Anycast-RP with MSDP, and with fewer protocols. In other words, Anycast-RP can still be used without the need to establish MSDP peering sessions between the RPs. The primary difference is that each RP will send copies of the Register messages to the other (peering) RPs that usually received them from the registered sources or the designated router (DR) that is directly attached to the multicast source LAN. Consequently, each RP can maintain a list of the sources of each multicast group registered with different RPs (S,G). As a result, the multicast streams from the different sources distributed across the PIM multicast domain can reach all receivers even if they are associated with different RPs.

As shown in [Figure 10-21](#)

1. The closest RP (anycast loopback IP) for each source and receiver will be selected by the underlying unicast routing protocol (IGP).
2. When RP-B receives the PIM Register message from multicast source S-B via R-1(DR), it will decapsulate it and then forward it across the shared tree toward the interested (joined) receivers.
3. Assuming all RPs (RP-A, RP-B, and RP-C) are deployed as Anycast-RP PIM among each other, in this case RP-B will send a copy of this Register message to RP-A and RP-C.
4. RP-A and RP-C will usually decapsulate the Register message sent from RP-B and in turn will forward it down to the shared tree (if there is any interested/joined receiver; otherwise, the RP can discard the packet).
5. If RP-B fails, IGP will route any multicast messages from the multicast source behind R-1 (S-B) toward the closest Anycast-RP (in this example, RP-C).

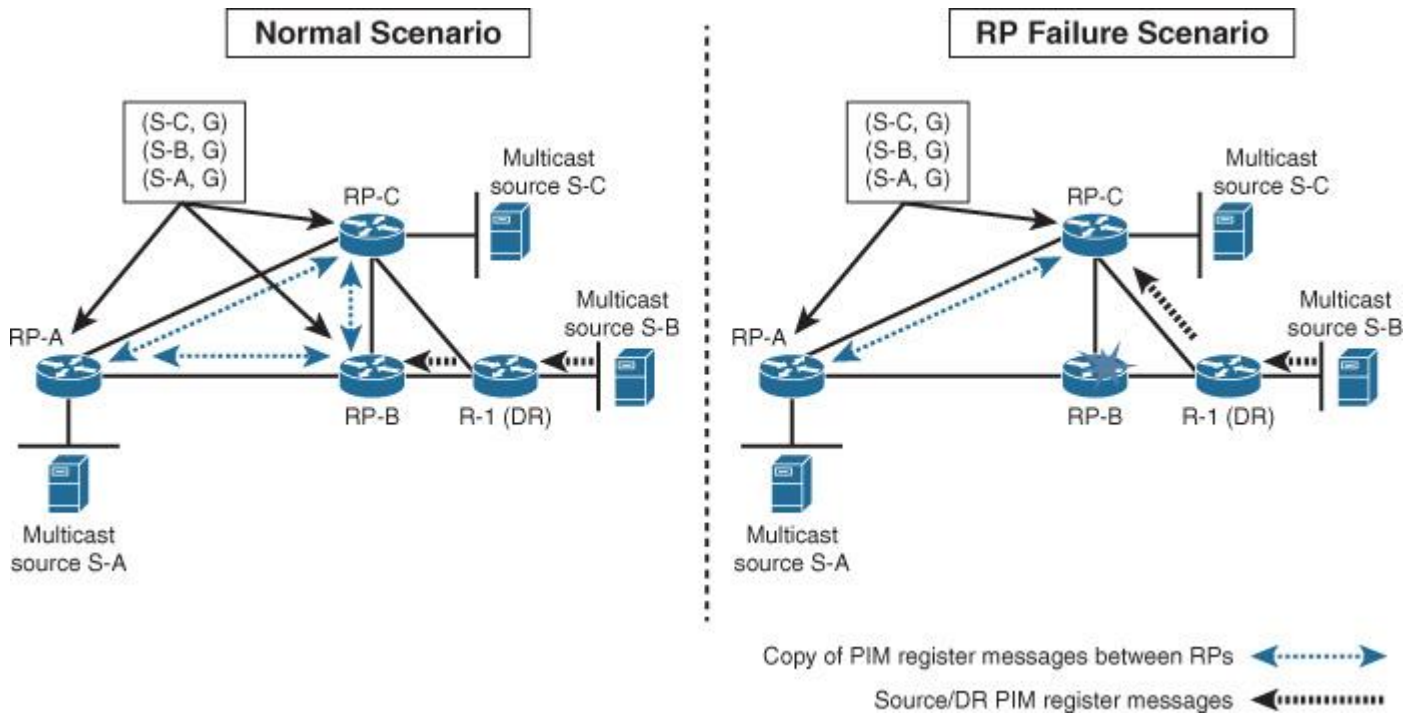


Figure 10-21 Anycast-RP PIM

However, MSDP may still be required in IPv4 interdomain multicast scenarios (for example, if the internal sources need to be advertised to multicast RPs outside of the local PIM domain, and similarly if learning of external multicast sources is required). In addition, MSDP rides on top of TCP, so it also offers guaranteed delivery. Also, from an operations and troubleshooting point of view, MSDP compared to Anycast-RP RFC 4610 offers higher flexibility because it has an SA-cache and multiple **show** commands that help network operators to know in a simplified way where it learned stuff from and what is in the cache.

Phantom RP

With PIM-Bidir, all the packets technically flow over the shared tree. Therefore, redundancy considerations of the RP become a critical requirement. The concept of phantom RP is specifically used for PIM-Bidir, and is not necessarily to be a physical RP/router. However, an IP subnet that is routable in the network can serve the purpose as well, where the shared tree can be rooted as shown in [Figure 10-22](#).

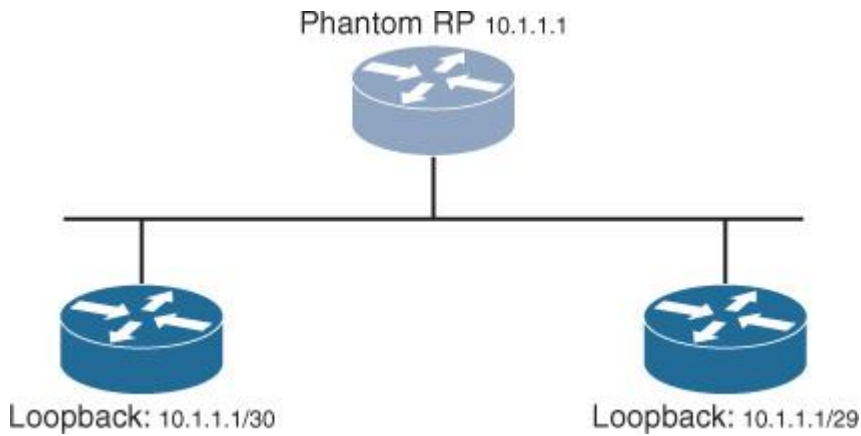


Figure 10-22 *Phantom RP*

As shown in [Figure 10-22](#), if you configure two routers in a network with the same IP and different subnet mask, IGP can control the preferred path for the root (phantom RP) of a multicast shared tree based on the longest match (longest subnet mask) where multicast traffic can flow through. The other router with the shorter mask can be used in the same manner if the primary router fails. This means the failover to the secondary shared tree path toward the phantom RP will rely on the unicast IGP convergence. Furthermore, if Auto-RP is used as a dynamic RP discovery mechanism to reduce operational overhead, a mapping agent is required for Auto-RP to work properly.

Live-Live Streaming

The term *live-live* refers to the concept of using two live simultaneous multicast streams through the network using either a path separation technique or a dedicated infrastructure and RPs per stream. For instance, as shown in [Figure 10-23](#) and [Figure 10-24](#), the first stream (A) is sent to one set of multicast groups, and the second copy of the stream (B) is sent using a second set of multicast groups. Each of these groups will usually be delivered using separate infrastructure equipment to the end user with complete physical path separation, as shown in [Figure 10-23](#). This design approach offers the ultimate level of resiliency that caters for any failure in a server or in network component along the path.

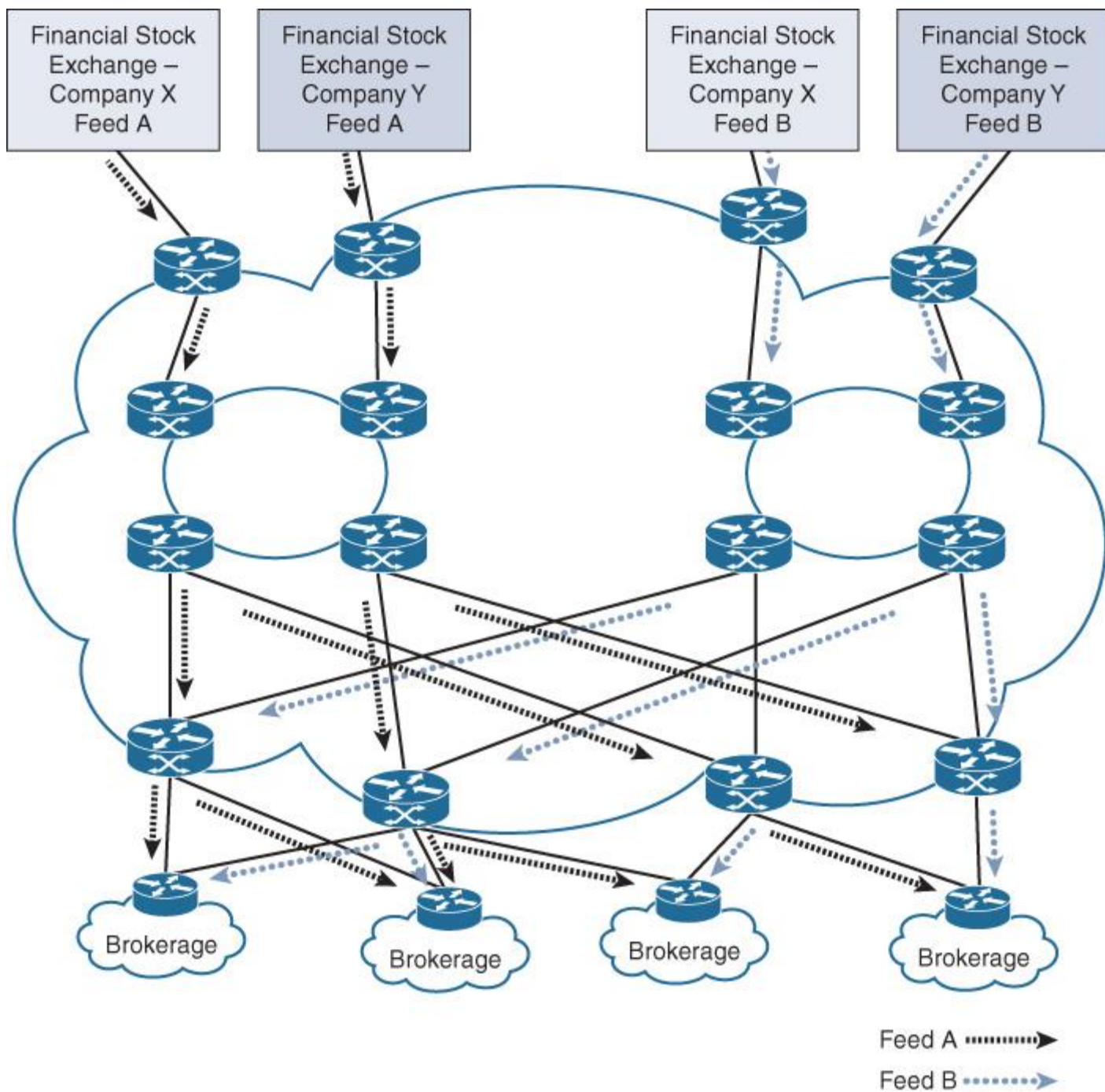


Figure 10-23 Live-Live Stream over Separate Core Networks

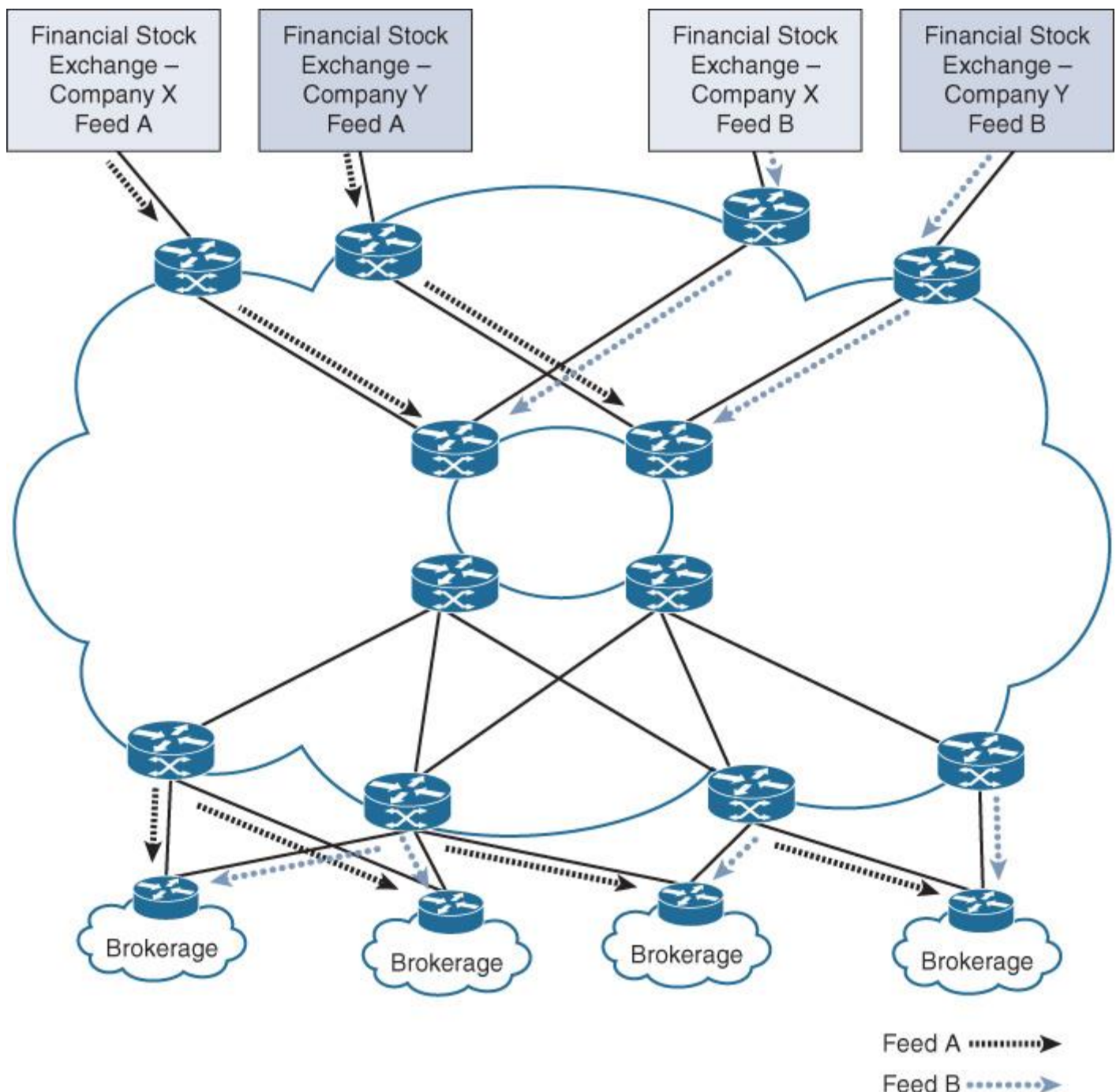


Figure 10-24 *Live-Live Stream over Single Core Network*

However, using single infrastructure such as MPLS-enabled core associated with MPLS-TE to steer the streams over different paths across the core infrastructure, as shown in [Figure 10-24](#), offers resiliency for any failure on the server side; however, it may not offer full network resiliency (because both streams will use the same core infrastructure). That said, if the MPLS provider catered for different failure scenarios (optical, node, link, and so on, along with switchover time that is fast enough to be performed without being noticed by the applications, such as using MPLS-TE FRR, and also avoids any shared risk link group [SRLG] along the path), in this case it can offer a reliable and cost-effective solution.

One of the primary drivers to adopt such an expensive design approach (live-live) is the strict requirement of some businesses (commonly in financial services industry because each millisecond is worth money) to aggressively minimize the loss of packets in the multicast data streams and a reliable and low-latency multicast solution that does not introduce retransmissions.¹¹

¹¹. "Market Data Network Architecture (MDNA) Overview," <http://www.cisco.com>

First Hop Redundancy Protocol-Aware PIM

In a shared LAN segment with a pair of Layer 3 gateways deployed with First Hop Redundancy Protocol (FHRP) such as Hot Standby Routing Protocol (HSRP) to provide first-hop redundancy, the PIM designated router (DR) election is unaware of the FHRP redundancy deployment. As a result, the HSRP active router (AR) and the PIM DR of the same LAN segment will probably not be the same router because PIM has no inherent redundancy capabilities and normally operates independently of HSRP.

HSRP-aware PIM helps network designers to ensure that multicast traffic is forwarded via the HSRP (AR) that will be acting as the PIM (DR) for the LAN segment as well. This will allow PIM to leverage HSRP redundancy and to offer failover capability derived from the HSRP states in the device. "With HSRP-Aware PIM enabled, PIM sends an additional PIM Hello message using the HSRP virtual IP addresses as the source address for each active HSRP group when a device becomes HSRP active."¹² Therefore, downstream devices such as a Layer 3 node or a multicast streaming server can use either a static route or a default gateway that points to the FHRP virtual IP (VIP), which offers a more reliable and simplified operation with simple configuration at the downstream node side, as shown in [Figure 10-25](#).

12. "HSRP Aware PIM," "VRRP Aware PIM," <http://www.cisco.com>

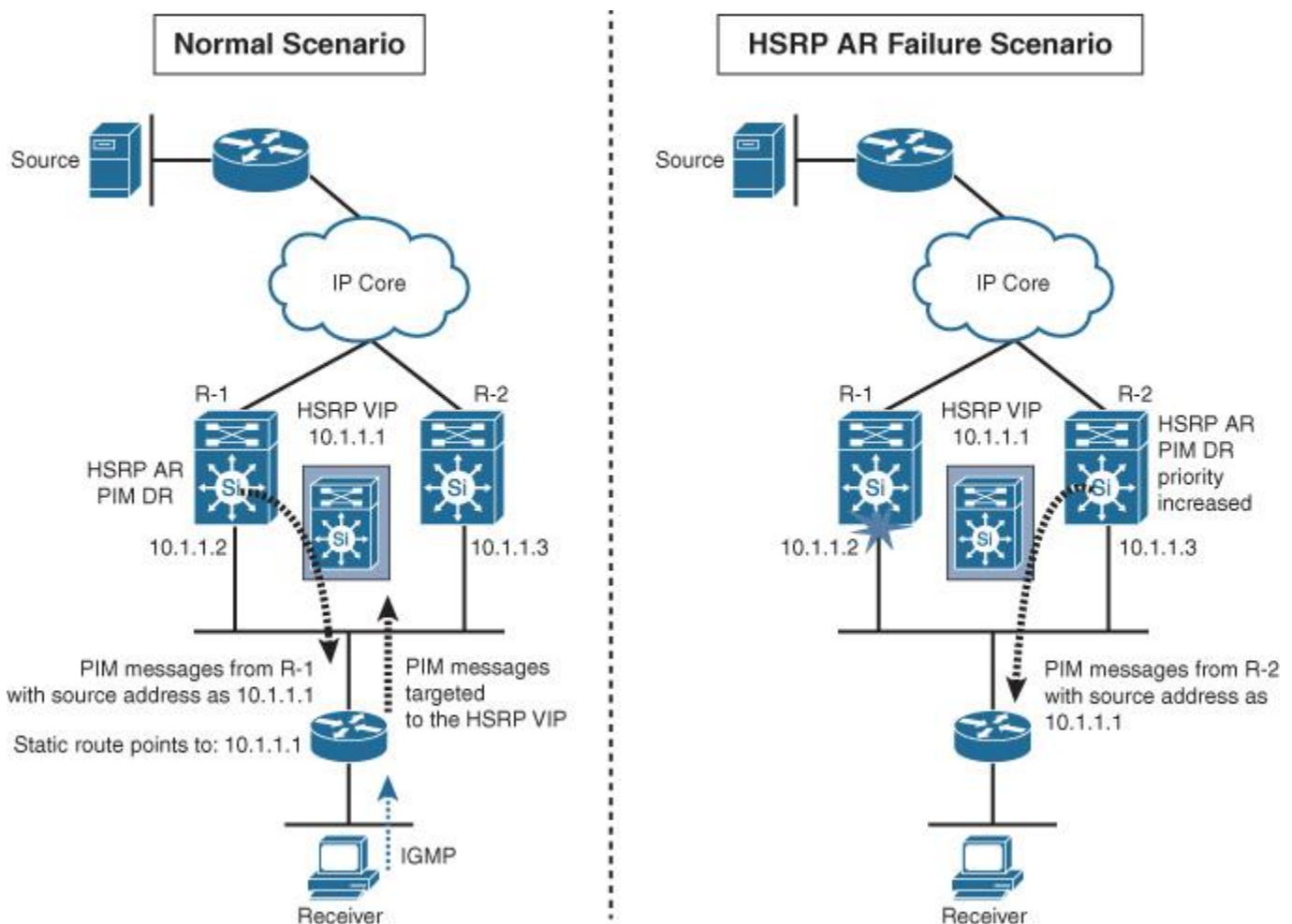


Figure 10-25 HSRP-Aware PIM

From a resiliency point of view, following an HSRP AR failover or recovery event, the newly elected HSRP AR will assume the responsibility for the routing and forwarding of all the traffic sent to the HSRP VIP. At the same time, PIM will tune the priority of the DR based on HSRP state to maintain the alignment of the HSRP AR and the multicast DR functions on the same node.

Final Thoughts on IP Multicast Design

Based on the multicast design options, considerations, and constraints covered in this section, network designers should always answer the following questions before considering any design recommendation or strategy:

- What is the targeted business or environment (enterprise versus SP)?
- Why is multicast required? Is it to run a business-critical application, to provide a transport network for customers to run their applications, or to provide a service, such as IPTV?
- What are the characteristics of the multicast application? Many to many versus one to many, IPv4 or IPv6? This can help to decide the type of RP and where the RP should be positioned in the network.
- Is the targeted network under single or multiple administrative domains? If multiple, is there any requirement to provide multicast communications between the different domains (interdomain)?
- What is the Layer 3 routing protocol in use, and what does the logical design look like? This helps to determine RPF check failure considerations.
- Is there any constraint related to enabling multicast across the network path? (For example, the WAN provider may not support multicast.)
- Does any network device in the path not support multicast, such as firewalls?
- Are there any staff knowledge limitations with regard to certain IP multicast protocols, such as LSM?
- Are there any high-availability requirements, such as redundant RPs?

Further Reading

“General Information About IP Multicast,” <http://www.cisco.com/go/multicast>

“Interdomain Multicast Solutions Using MSDP,” <http://www.cisco.com>

RFC 2117, *Protocol Independent Multicast-Sparse Mode (PIM-SM): Protocol Specification*, <http://www.ietf.org>

“Bidirectional PIM Deployment Guide,” <http://www.cisco.com>

RFC 6826, *Multipoint LDP In-Band Signaling for Point-to-Multipoint and Multipoint-to-Multipoint Label Switched Paths*, <http://www.ietf.org>

Multicast Only Fast Re-Route, draft-ietf-rtgwg-mofrr-06, <http://www.ietf.org>

QoS Design Considerations

In today’s converged networks, there is an extremely high reliance on IT services and applications. In particular, there is increased demand on real-time multimedia applications and collaboration services (also known as *media applications*) over one unified IP network that carries various types of traffic streams such as voice, video, and data applications. For instance, voice streams can be found across the network in different flavors, such as standard IP telephony, high-definition audio, and Internet Voice over IP (VoIP). In addition, video communications also have various types, where each has different network requirements, such as video on-demand, low-definition interactive video (such as webcams), high-definition interactive video (such as telepresence), IP video surveillance, digital signage, and entertainment-oriented video applications [94]. However, there can be an unlimited number of data applications. (See [Table 2-1, Chapter 2, “Enterprise Layer 2 and Layer 3 Design,”](#) for an example of the various applications of today’s converged networks.)

Therefore, to deliver the desired quality of experience to the different end users (as discussed in [Chapter 1, “Network Design Requirements: Analysis and Design Principles”](#)), network designers and architects need to consider a mechanism that can offer the ability to prioritize traffic selectively (usually real-time and mission-critical applications) by providing dedicated bandwidth, controlled jitter and latency (based on the application requirements), and improved loss characteristics, and at the same time ensuring that providing priority for any traffic flow will not make other flows fail. This mechanism is referred to as *quality of*

service (QoS). The following sections discuss the design approaches and considerations of QoS using a business-driven approach.

QoS High Level Design: Business-Driven Approach

To design and deploy QoS successfully across the network, in particular converged networks, network designers or architects need to consider the top-down design approach to initially understand the critical and important applications from the business point of view. Then, they can assess the optimal QoS design strategies to meet the business and application requirements.

As discussed earlier, the goal of the network is to become a business enabler. To achieve that, the network ideally should act as a service delivery mechanism that facilitates the achievement of the business's goals. Therefore, it is critical to align QoS design with business priorities, expectations, and requirements to deliver the desired level of quality of experience.

For instance, an organization may have a financial application that is sensitive to packet loss, and any loss of connectivity can cost the business a large amount of money. Accordingly, this application must be treated as a high-priority application. Similarly, if an SP needs to meet a very strict SLA with its customers to deliver their voice traffic with no more than 1 percent of end-to-end packet loss, this SP must consider a suitable QoS strategy and apply the right QoS design to meet these SLA requirements. Otherwise, it will lead to a business loss. This loss can be tangible, such as a penalty, intangible, such as reputation, or most probably both. Therefore, network designers must have the right QoS design approach and strategy to avoid the high volume of application complexities in today's networks and provide the desired level of quality of experience to the business and end users. [Table 10-11](#) provides a summarized (top-down) QoS design approach and strategy (in order).

Strategic Goal	Approach	Design Considerations
Understand business requirements.	Understand business priorities and goals.	Identify primary business drivers. Highlight the constraints (for example, budget).
Identify the scope.	Understand the scope of the QoS design such as campus, WAN, VPN or SP edge or end to end across different blocks.	Is the application used within the campus, across the WAN, or over VPN? Is there any network in the path not directly controlled such as WAN?
Identify mission-critical applications.	Identify which application need to be treated differently (prioritized). Identify which application is a nonbusiness application (deprioritize).	What are the mission-critical applications or services (for example, SAP, FCoE, VoIP, TelePresence)

Understand application requirements.	Identify the characteristic of each application.	What sort of network delivery is required: TCP, UDP, unicast multicast, and so on? Application sensitivity to pack loss, jitter, delay.
Select a design strategy and identify the technical constraints.	Clarify the end-to-end design strategy in terms number of QoS classes, QoS toolset to be used, and so on.	What traffic classification strategy is to be used within the LAN (for example 8 or 12 classes)? What MPLS DiffServ tunneling mode is used? Is the core/WAN underlay native IP or MPLS based? What are the classes of service (CoS) supported over the WAN (for example, MPLS provider)? Can the targeted network node support the required number of queues? Or does it support any priority queuing (technical constraints)?

Table 10-11 *Top-Down QoS Design Approach Summary*

QoS Architecture

In general, there are two fundamental QoS architecture models:

■ **Integrated services (IntServ):** This model, as specified in RFC 1633, offers an end-to-end QoS architecture based on application transport requirements (usually per flow) by explicitly controlling network resources and reserving the required amount of bandwidth (end to end along the path per network node) for each traffic flow. Resource reservation protocols, such as RSVP, and admission control mechanisms form the foundation of this process.

■ **Differentiated services (DiffServ):** This model, specified in RFC 2475, offers a QoS architecture based on classifying traffic into multiple subclasses where packets flows are assigned different markings to receive different forwarding treatment (per-hop behavior, or PHB) per network node along the network path within each differentiated services domain (DS domain).

Note

The aforementioned QoS architectural models are applicable for both IPv4 and IPv6, because both IP versions include the same 8-bit field in their headers, which can be used, for example, for DiffServ (IPv4: Type of Service [ToS]; IPv6: Traffic Class). Therefore, the concepts and methodologies discussed in this section are intended for both IPv4 and IPv6 unless otherwise specified, such as if an application supports and uses the added 20-bit Flow Label field of the IPv6 header (RFC 2460, 3697). However, the larger IPv6 packet's header needs to be considered when calculating the aggregate bandwidth of traffic flows.

QoS DiffServ Architecture and Toolset

A true and effective QoS design must cover traffic flows of applications and services end to end. This can sometimes be complicated because each single traffic flow traverses multiple networks and is architected with a different QoS philosophy. Therefore, for network architects and designers to create such a cohesive

design model, the design must be divided into domains commonly called and as described in RFC 2475 as *differentiated services domains* (DS domains). Each DS domain usually consists of multiple interconnected network nodes (such as switches and routers) operating under a common service provisioning policy, along with set of PHB groups enabled on each node that offers network designers a framework to facilitate the design of scalable and flexible DS domains.

Note

Per-hop behavior (PHB) defines a forwarding behavior with regard to the treatment characteristics for the traffic class, such as its dropping probabilities and queuing priorities, to offer different levels of forwarding for IP packets (described in RFC 2474). DS domains described in this section refer to any domain with QoS policies and differentiated treatments regardless if it is IP Precedence based (RFC 791), Assured Forwarding based (RFC 2597), or a mixture of both.

Note

Service provisioning policy refers to the attributes of traffic conditioners (QoS policies) deployed at each DS—domain boundary and mapped across the DS domain.

Each single DS domain can be divided into two primary types of DS network nodes. The first type is internal, which usually refers to the nodes belonging to a single DS domain. As described earlier, these nodes should have the same QoS or DS provisioning policy. The second type of DS node is the DS boundary node, which faces other DS or non-DS-capable domains. Typically, these nodes are the ones that must be responsible for applying traffic polices (QoS policies) on traffic flows in both directions (ingress and egress) based on a predefined or agreed model between the DS domains that is ideally driven by traffic and application requirements, as shown in [Figure 10-26](#).

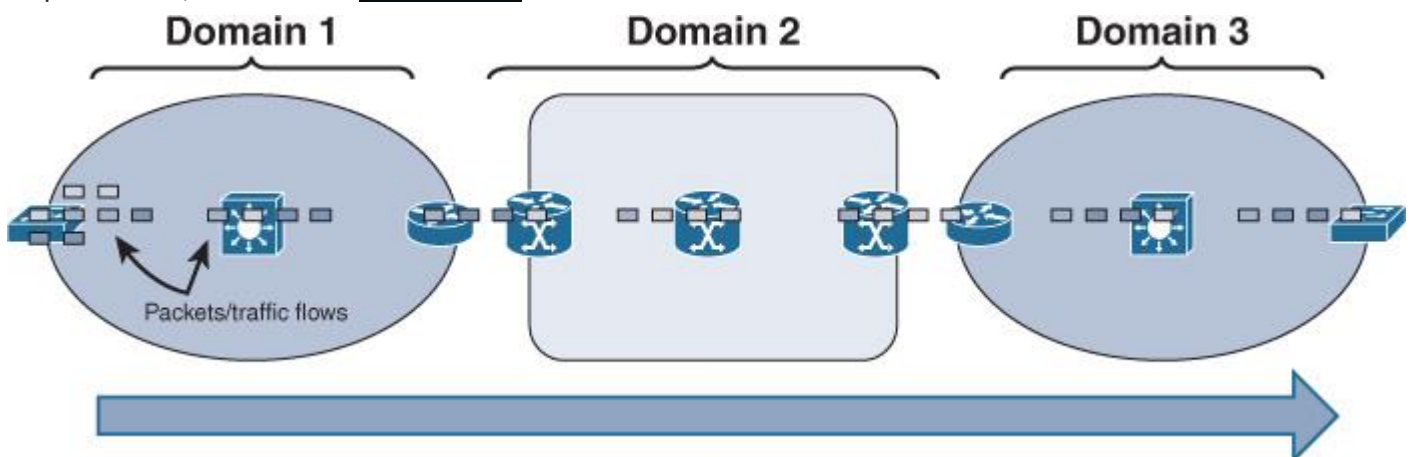


Figure 10-26 QoS DS-Domains

Practically, DS domains can take different forms, such as the following:

- Enterprise domain with a SP domain in the middle (WAN transport).
- Within an enterprise, there can be multiple DS domains: campus LAN, WAN, DC, and DMVPN over Internet.

The second scenario consists of multiple DS domains, which all belong to a single administrative authority and that can be combined under one global DS domain or region. Each DS domain in that region has the flexibility to have its own QoS provisioning standards to offer more structured and tiered QoS domains design for large-scale networks with a large number of distributed blocks, as shown in [Figure 10-27](#).

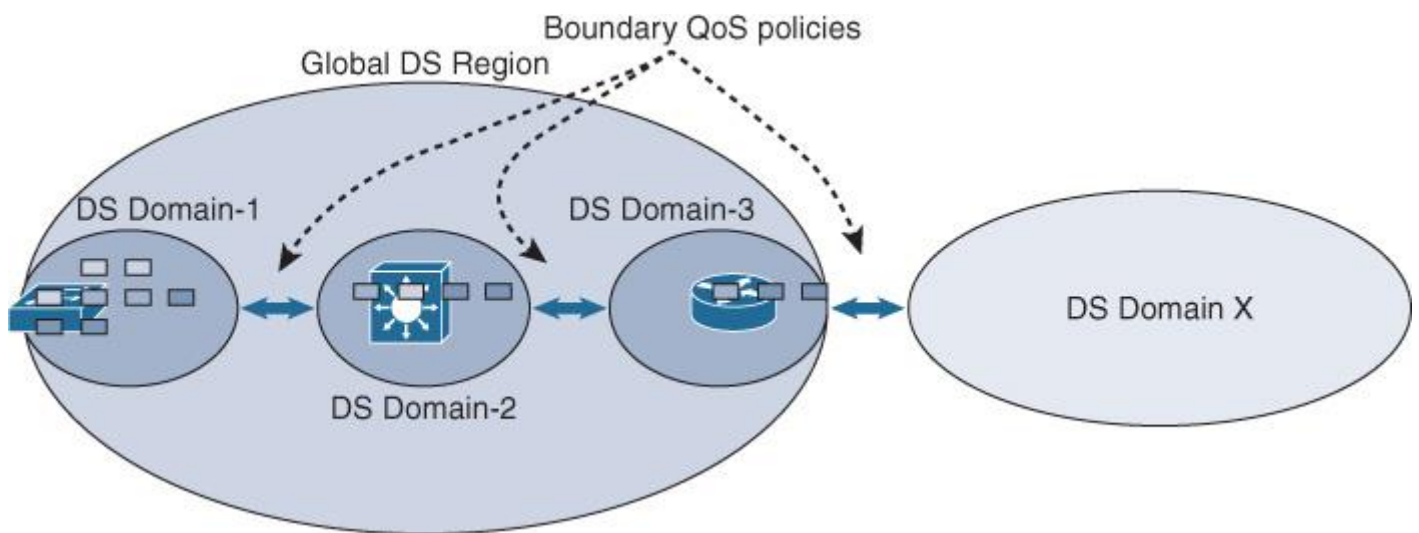


Figure 10-27 Multitier DS Domains

Each domain is usually under a single administrative and control authority. Hypothetically, each authority defines cohesive end-to-end, measurable, and quantifiable attributes per DS subdomain and per global DS domain, which should be driven by the characteristics and requirements of the services or applications running across this domain (traffic flow aggregates). Typically, there can be multiple possible points across each QoS domain specifying where policies and traffic treatment can be enforced to influence the experience of packets as they enter, cross, and exit a DS domain. Traffic conditioning and QoS policies are the key QoS elements that serve this purpose within each domain and between the different domains using the following primary QoS toolset:

- Traffic classification and marking
- Traffic profiling and congestion management
- Congestion avoidance (active queue management)
- Admission control

The subsequent subsections cover these QoS toolset in more detail.

Traffic Classification and Marking

Classification refers to the process of selecting frames or packets in a traffic stream based on the content of some portion of the frame or packet header to which different policies can then be applied. Traffic marking, on the other hand, writes a value into the packet header to be identified by QoS policies and placed in the desired class with the desired treatment at different stages during the end-to-end packet trip from the source to the intended destination (within and across DS domains).

However, classification of traffic does not always require marking to be applied. For instance, in some scenarios, traffic only needs to be selected based on the value of a combination of one or more IP header fields, such as source address, destination address, source port, destination port numbers, or an incoming interface then to be associated with a QoS policy action, such as placing it in a predefined QoS queue. Classification of traffic almost always should be performed at the point of network access (as close to the traffic source as possible), and then to be associated with the appropriate marking value (usually ToS header bits) in which network designers can have the flexibility to select this traffic and apply the desired QoS policies at any node across the network (usually within a single DS domain or region).

In addition, marking can establish trust boundaries at the edge of the network. A *trust boundary* refers to the point within the network where markings such as CoS or differentiated services code point (DSCP) begin to be accepted as its set by the connected node or endpoint, such as an IP phone that set its voice traffic with a DSCP-PHB value of EF. However, not every endpoint can mark its traffic. Therefore, trust boundaries are commonly classified into three primary models: trusted, untrusted, and conditional trust, as shown in [Figure 10-28](#).

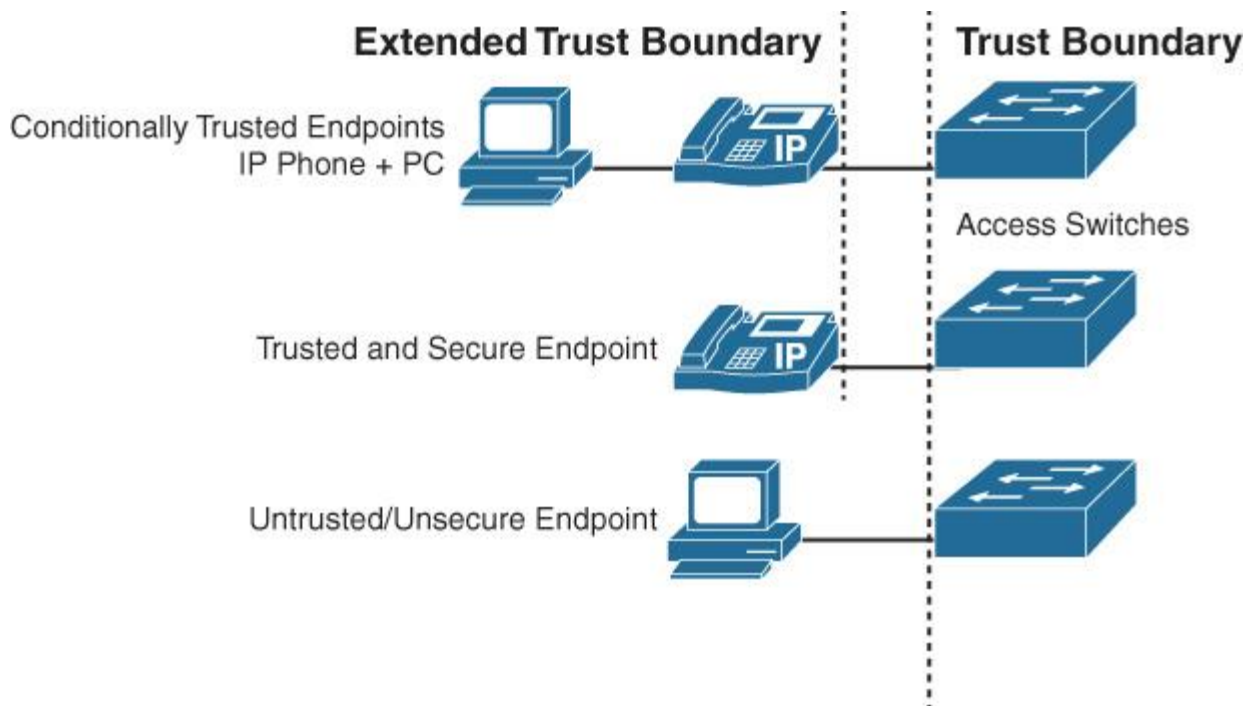


Figure 10-28 QoS Trust Boundaries

■ **Trusted model:** This model can be used with endpoints that can mark their traffic. At the same time, these endpoints have to be approved and trusted from a security point of view, such as IP phone, voice gateways, wireless access points, videoconferencing, and video surveillance endpoints. In addition, “ideally” these trusted endpoints should not be mobile (instead fixed endpoints) to be reflected at the switch port level in a more controlled manner.

■ **Untrusted model:** This model usually considers using a manual traffic classification and marking. The most common candidates of this model are PCs and servers, because these endpoints are subject to attack and infection by worms and viruses that can flood the network with a high volume of malicious traffic. Even worse, this traffic might be marked with a CoS or DSCP value that has priority across the network, which usually leads to a true denial-of-service (DoS) situation. However, PCs and servers normally run business-critical applications that need to be given certain service differentiation across the network, such as a PC running a software-based phone or server running business-critical applications such as SAP or CRM. With this model network, designers can selectively classify each of the desired application’s traffic flows to be treated differently across the network and mark it with the desired CoS/DSCP value along with a policy that either limits each application class to a predefined maximum bandwidth or marks down the out-of-profile traffic to a CoS/DSCP value that has lower priority across the network. Furthermore, as a simple rule of thumb, any endpoint that is not under control by the enterprise should be considered as untrusted, and the classification and marking of traffic flows can be controlled selectively and manually.

■ **Conditional trust model:** This model offers the ability to extend the trust boundary of the network edge to the device or endpoint connected to. This is based on an intelligent detection of a trusted endpoint, usually an IP phone. (In Cisco solutions, this is achieved by using Cisco Discovery Protocol [CDP].) However, the IP phone in this scenario has a PC connected to the back of the IP phone. Therefore, by extending the trust boundary to the IP phone, the IP phone can send its traffic in a trusted manner while overriding PC traffic to DSCP value of 0. This model offers a simple and easy method to roll out large IP telephony deployments with minimal operational and configuration complexity. However, if the PCs have some applications that need to be marked with a certain DSCP value, such as a softphone or a business video application, in this case manual traffic classification and marking is required at the edge port of the access switch to identify this traffic and mark it with the appropriate CoS/DSCP value and ideally associate it with a policer.

Furthermore, marking values can be changed (re-marked) based on the QoS design requirements at any location within a DS domain or between DS domains. The most common example is that between DS

domains there might be a mismatch between the ToS values used to classify traffic. Therefore, at the egress or ingress, re-marking has to take place to maintain an end-to-end unified QoS model. In contrast, in a single DS domain, re-marking is commonly used to move traffic that is out of profile (out of policy limit) into a QoS class with lower bandwidth or priority as a protective countermeasure by re-marking the out-of-profile traffic flows with a different ToS value at any location within a the DS domain. [Table 10-12](#) summarizes the most common classification and marking options.

OSI Layer	Classification	Marking
Physical layer	Input interface	N/A
Layer 2	VLAN ID, MAC, IEEE 802.1Q/p CoS	IEEE 802.1Q/p CoS
Layer 2.5	MPLS label, MPLS EXP	MPLS EXP
Layer 3	IP DSCP, IP based (such as IP source/destination)	IP (IPP, DSCP)
Layer 4	Port-based source/destination	IPP, DSCP, EXP
Higher layers up to 7	Application signature such as using a network-based application recognition (NBAR) framework	IPP, DSCP, EXP

Table 10-12 *Summary of QoS Classification and Marking Options*

Note

DSCP marking is more commonly used than IP Precedence (IPP) because of its higher flexibility and scalability to support a wider range of classes. However, in some scenarios, a mix of both may be required to be maintained, such as migration or integration between different domains (such as in merger and acquisition scenarios and WAN MPLS VPN SPs that offer their CoS classes based on IPP) where seamless QoS interoperability has to be maintained. In this case, class selector PHB is normally used to provide backward compatibility with ToS-based IP Precedence (RFC 4594, 2474).

Logically, after traffic flows are classified and marked (whether manually at the edge of the DS domain or automatically by being considered one of the trust boundary models discussed earlier), traffic flows have to be grouped in DS classes. Usually, application flows that share similar or the same traffic characteristics and network requirements such as delay, jitter, and packet loss can be placed under the same DS class to achieve a structured traffic grouping that helps network operators to assign the desired treatment at different locations across the DS domain (per class), such as assigning different queuing models per class to control traffic flows during periods of traffic congestion.

Traffic Profiling and Congestion Management

During normal situations where traffic passing through the network is under or equal to the actual maximum available bandwidth capacity of its links, packets are usually sent out of the interface as soon as they arrive. In contrast, in traffic congestion situations, packets normally arrive faster than the outgoing interface can handle them. This will lead to undesirable outcomes to traffic flows that most probably will impact the business-critical application's performance and users' quality of experience. In other words, technically, if the network is overprovisioned with bandwidth, there may be no need for QoS considerations, because it will add minimal value. However, it is a common practice for QoS to be enabled with a minimum number of classes to cater for critical applications in case of unpredicted congestion that might occur, such as a failure of the upstream network node, which may lead to overutilization of the secondary path if the capacity planning did not consider different failure scenarios.

Therefore, to cater for network congestion situations, a network designer must plan for an effective mechanism to be used to manage traffic during congestion periods. This is commonly referred to as *congestion management*. When congestion management is used across the network, nodes can queue the accumulating packets at the outbound interface until the interface (Tx-Ring) is free to send the queued packets. The transmission of the queued packets, however, is scheduled based on the assigned desired priority and queuing mechanism configured at the interface level per traffic flow aggregate (predefined traffic profiling).

In fact, with congestion management, placing packets in a predefined transmission queue is one half of the job. The other half of it is how the different queues are serviced with respect to each other during a congestion period at the router/interface level. [Table 10-13](#) summarizes the common queuing mechanisms that can be used for the purpose of congestion management, along with the characteristics of each.

	Weighted Fair Queuing (WFQ)	Priority Queuing (PQ)	Class-Based Weighted Fair Queuing (CBWFQ)
Characteristic	The WFQ algorithm offers a dynamic fair distribution among all traffic flows based on weight.	Typically supports four queues with different priority levels, and the higher-priority queues are always serviced first	Provides class-based queuing (user-defined classes) with minimum bandwidth guarantee. Supports flow-based WFQ for nondefined classes, such as class- default. Supports low-latency queuing (LQQ)

Table 10-13 Common QoS Queuing

Note

There other queuing techniques not covered in this section because they are less commonly used or offer basic queuing capabilities. However, this does not mean that they are not used or cannot be considered as an option. These other techniques include weighted round-robin (WRR) and custom queuing. First-in, first-out (FIFO) queuing on is the default queuing when no other queuing is used. Although FIFO is considered suitable for large links that have low delay with very minimal congestion, it has no priority or classes of traffic.

Although WFQ offer a simplified, automated, and fair flow distribution, it can impact some applications. For instance, a telepresence endpoint may require end-to-end 5 Mbps for the video Rapid Transport Protocol (RTP) media stream over a 10-Mbps WAN link, and there may be multiple flows passing over the same link, let’s say ten flows in total. Typically, with WFQ fairness, telepresence video streams will probably get one-tenth of the total available bandwidth of the 10 Mbps, which will lead to a degraded video quality. With CBWFQ, though, network designers can place the flows of the telepresence RTP media streams in their own class with a minimum bandwidth guarantee of 5 Mbps during congestion periods. In addition, interactive video traffic can be assigned to the LLQ to be prioritized and serviced first during an interface congestion situation, as shown in [Figure 10-29](#).

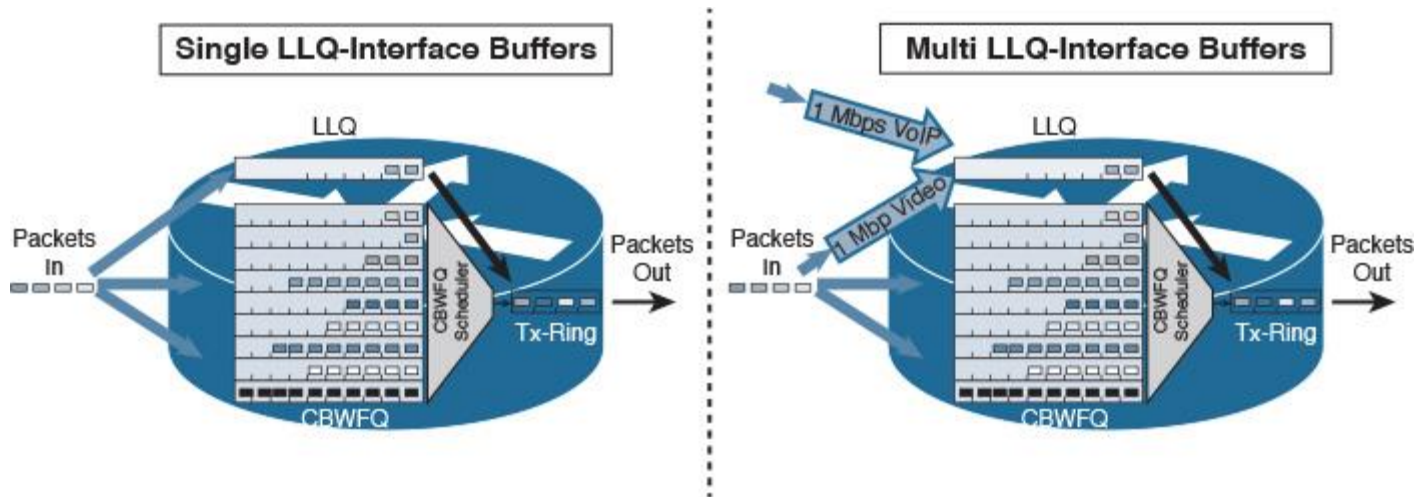


Figure 10-29 CBWFQ with LLQ

As shown in [Figure 10-29](#), CBWFQ supports two models of LLQ: single LLQ and the multi-LLQ. Typically, with the multi-LLQ model, there will be a single aggregate LLQ from the software point of view, and multiple sub-LLQs can be enabled inside this strict priority queue. With this model network, operators can assign multiple traffic flow types to the LLQ, such as VoIP and video. However, inside the LLQ itself, the service is based on the FIFO concept. Therefore, within each parent LLQ admission control, it is required to protect LLQ from another LLQ, such as protecting a voice LLQ from a video LLQ.

Note

The different Cisco software versions, such as IOS, include a built-in policer (implicit policer) with the LLQ, which limits the available bandwidth of the LLQ (such as real-time traffic flows) to match the bandwidth allocated to the strict-priority queue, thus preventing bandwidth starvation of the nonreal-time flows serviced by the CBWFQ scheduler. However, this behavior (implicit LLQ policing) is applicable only during periods of interface congestion (full Tx-Ring). A similar concept is applicable to the multi-LLQ model, where a state implicit policer is enabled per LLQ [\[83\]](#).

Hierarchical QoS

The most common scenario at the enterprise edge is that links are provisioned with sub-line rate. For instance, the WAN SP may provision the physical connectivity over a 1-Gbps Ethernet copper link, whereas the actual provisioned bandwidth can be 10 Mbps, 50 Mbps, or any sub-line rate. The problem with this setup, even if QoS policies are applied on this interface, such as CBWFQ, is that it will not provide any value or optimization because QoS policies normally kick in when the interface experiences congestion. In other words, because the physical link line-rate in this scenario is higher than the actual provisioned bandwidth, there will be no congestion detected even if the actual provisioned sub-line rate is experiencing congestion, which means that QoS has no impact here. Therefore, with hierarchical QoS (HQoS), the shaper at the parent policy (as shown in [Figure 10-30](#)) can simulate what is known as *back pressure* to inform the router that congestion has occurred in which QoS policies can take place.

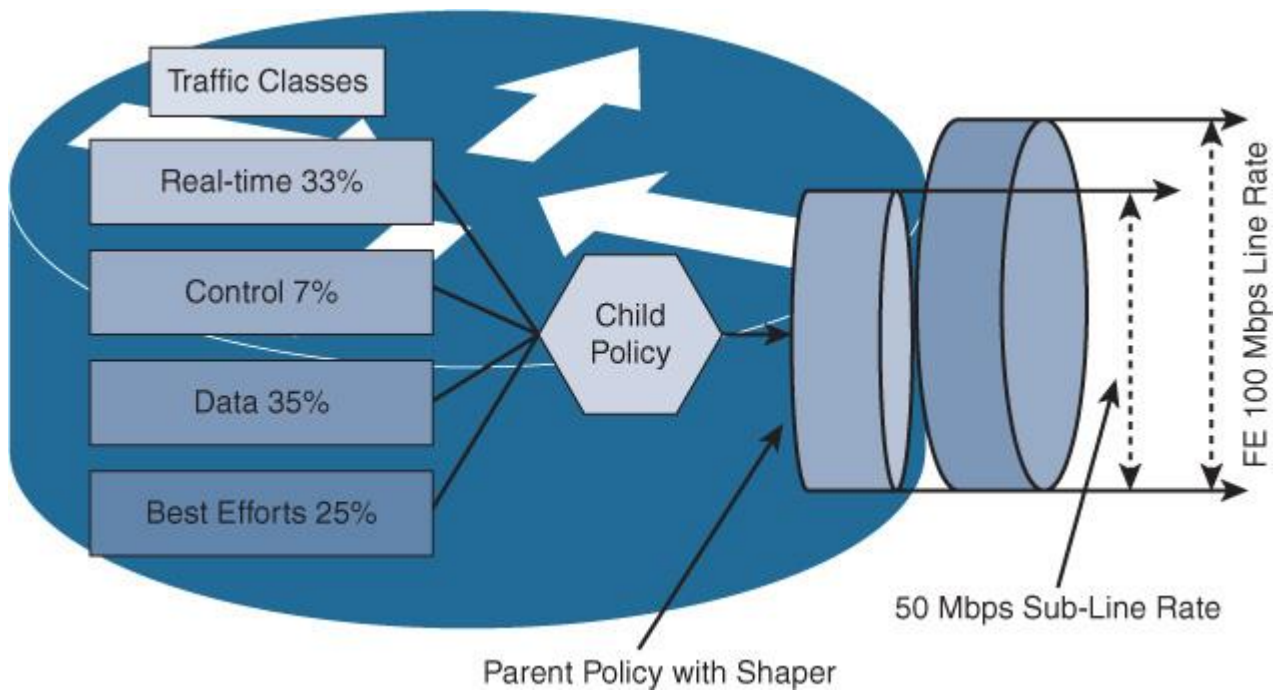


Figure 10-30 HQoS

Congestion Avoidance (Active Queue Management)

The congestion management techniques discussed in the previous section help network designers and operators to manage the front of the queue in terms of which packets should be sent out first. Congestion avoidance algorithms, in contrast, manage the tail of the queue in terms of which packets should be dropped first when queuing buffers are full. One of most commonly used and proven techniques here is the weighted random early detect (WRED), where packets are dropped based on their ToS markings, either IP Precedence based (where packets with lower IPP values are dropped more aggressively than higher IPP values) or DSCP based (where higher AF drop precedence values are dropped more aggressively). When the WRED algorithm starts selectively dropping packets, it will usually impact TCP windowing mechanisms to adjust the rate of flows to manageable rates. Therefore, WRED helps to optimize TCP-based applications [83].

Admission Control

Admission control is a common and essential mechanism used to keep traffic flows in compliance with the DS domain traffic conditioning standards, such as an SLA between a SP and its customers that specifies the maximum allowed traffic rate per class and in total per link, where excess packets will be discarded to keep traffic flows within the agreed traffic profile (SLA). There are two primary ways that admission control can be performed: traffic policing and shaping. With traffic policing, when traffic streams reach the predefined maximum contracted rate, excess traffic is either dropped or re-marked (marked-down), as shown in [Figure 10-31](#).

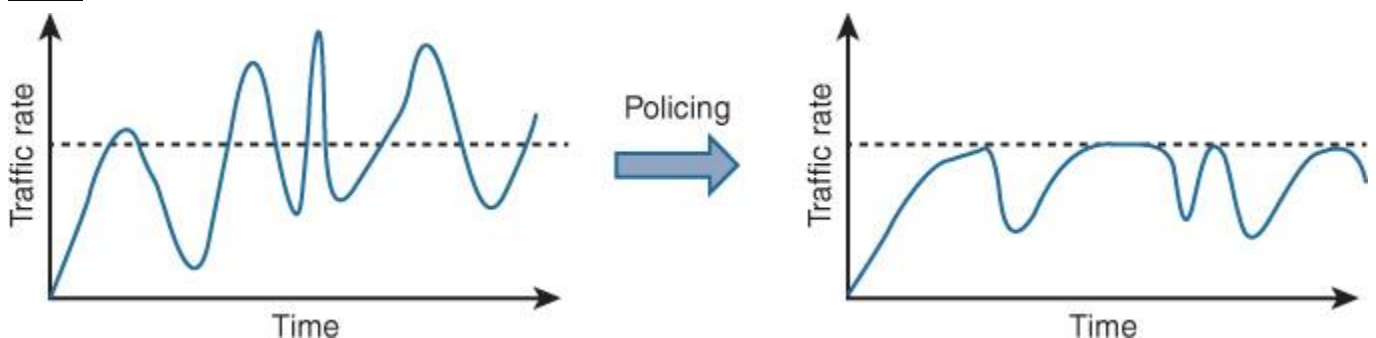


Figure 10-31 Traffic Policing

Traffic shaping, in contrast, keeps excess packets in a queue, buffered and delayed, and then schedules the excess for later transmission over increments of time. As a result, traffic shaping will smooth packet output rate and prevent unnecessary drops, as shown in [Figure 10-32](#). However, the buffering of excess packets may introduce delay to traffic flows, especially with deep queues. Therefore, with real-time traffic it is sometimes preferred to police and drop excess packets rather than delay it and then transmit it, to avoid degraded quality of experience.

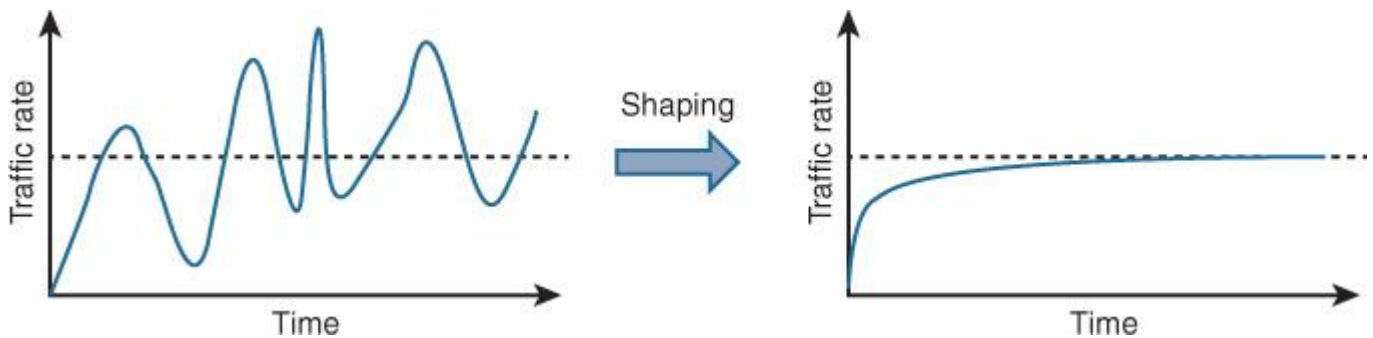


Figure 10-32 *Traffic Shaping*

QoS Design Strategy

Effective QoS design must always be measured end to end with regard to the packet's trip or traffic flow aggregates across the network (from the source to the intended destination). Otherwise, QoS will not deliver its value back to the business and end users. Therefore, it is important that network designers consider a consistent and unified QoS design model based on the different requirements, taking into consideration the underlying network infrastructures in terms of available bandwidth, traffic characteristics, and scope of the network, such as campus only versus WAN only or end to end across the entire enterprise (which translates to single DS domain versus multiple DS domains).

One of the critical elements of QoS design is traffic classification, marking, and profiling using a unified end-to-end approach. In particular, the strategy of marking IP traffic with a ToS value and mapping each type of traffic to a predefined QoS class will vary based on different factors. However, [Figure 10.33](#) provides a generic summarized recommendation of a QoS design model (strategy) that can serve as a baseline. This 12-class model is based on both Cisco's QoS Baseline and the informational RFC 4594. One of the main benefits of this model is that it provides a common and unified traffic marking and profiling characteristic across single and multiple DS domains, to a large extent.

Application Class	Per-Hop Behavior	IETF RFC	Queuing & Dropping	Application Examples
VoIP Telephony	EF	3246	Priority Queue (PQ)	IP Telephony (IPT)
Broadcast Video	CS5	2474	(Optional) PQ	IP Video Surveillance/IPTV
Real-time Interactive VC	CS4	2474	(Optional) PQ	Telepresence
Multimedia Conferencing	AF4	2597	BW Queue + DSCP WRED	IPT Video
Multimedia Streaming	AF3	2597	BW Queue + DSCP WRED	Video on Demand (VoD), E-learning
Network Control	CS6	2474	BW Queue	EIGRP, OSPF, BGP, HSRP, IKE
Call-Signaling	CS3	2474	BW Queue	SCCP, SIP, H.323
Mgmt (OAM)	CS2	2474	BW Queue	SNMP, SSH, Syslog
Low-Latency Data	AF2	2597	BW Queue + DSCP WRED	ERP Apps, CRM Apps, Database Apps
High-Throughput Data	AF1	2474	BW Queue + DSCP WRED	E-mail, FTP, Backup Apps, Content Distribution
Best Effort	DF		Default Queue + RED	Default Class
Low-Priority Data	CS1	3662	Min BW Queue (Deferential)	YouTube, iTunes, BitTorrent, Xbox Live

Figure 10-33 Twelve-Class QoS Baseline Model Based on Cisco and RFC 4594 Baselines

Note

While the IETF DiffServ RFCs provide a consistent set of PHBs for applications marked to specific DSCP values, they do not specify which application should be marked with which DSCP value. Therefore, considerable industry disparity exists in application-to-DSCP associations, which led Cisco to put forward a standards-based application marking recommendation in their strategic architectural QoS Baseline document (in 2002). Eleven different application classes were examined and extensively profiled and then matched to their optimal RFC-defined PHBs. More than four years after Cisco put forward its QoS Baseline document, RFC 4594 was formally accepted as an informational RFC (in August 2006). RFC 4594 puts forward 12 application classes and matches these to RFC-defined PHBs [96], as summarized in [Figure 10-33](#).

Note

The most significant of the differences between Cisco's QoS Baseline and RFC 4594 is the RFC 4594 recommendation to mark call signaling to CS5. Cisco has completed a lengthy and expensive marking migration for call signaling from AF31 to CS3 (as per the original QoS Baseline of 2002). It is important to remember that RFC 4594 is an informational RFC; in other words, it is only an industry best practice and not a standard [96].

The twelve-class QoS model is a comprehensive and flexible model, which can be standardized and considered across the enterprise network. However, this model is not always viable or achievable for the following several reasons:

- Not all enterprises or SPs are ready or need to deploy such a wide QoS design model.
- This 12-class QoS design can introduce a level of end-to-end QoS design and operational complexity, because most WAN providers offer either 4- or 6-class QoS models across the WAN.

Note

The biggest concern with regard to the operational complexity is the issues caused by human errors; this point is covered later in the “[Network Management](#)” section.

Ideally, what drives the number of classes (QoS model) is the applications used across the network that need special consideration and the level of service differentiation required for delivering the desired level of quality of experience. As shown in [Figure 10-34](#), if the WAN provider is offering a 4-class QoS model, it is not an easy task to map from the 12-class model to the 4 classes (outbound), and from 4-class model to the 12-class model (inbound) at each WAN edge, especially if there is a large number of sites. In contrast, if the enterprise is using a 4-, 6-, or 8-class QoS model, the operation and design complexities will be minimized with regard to QoS polices and configurations.

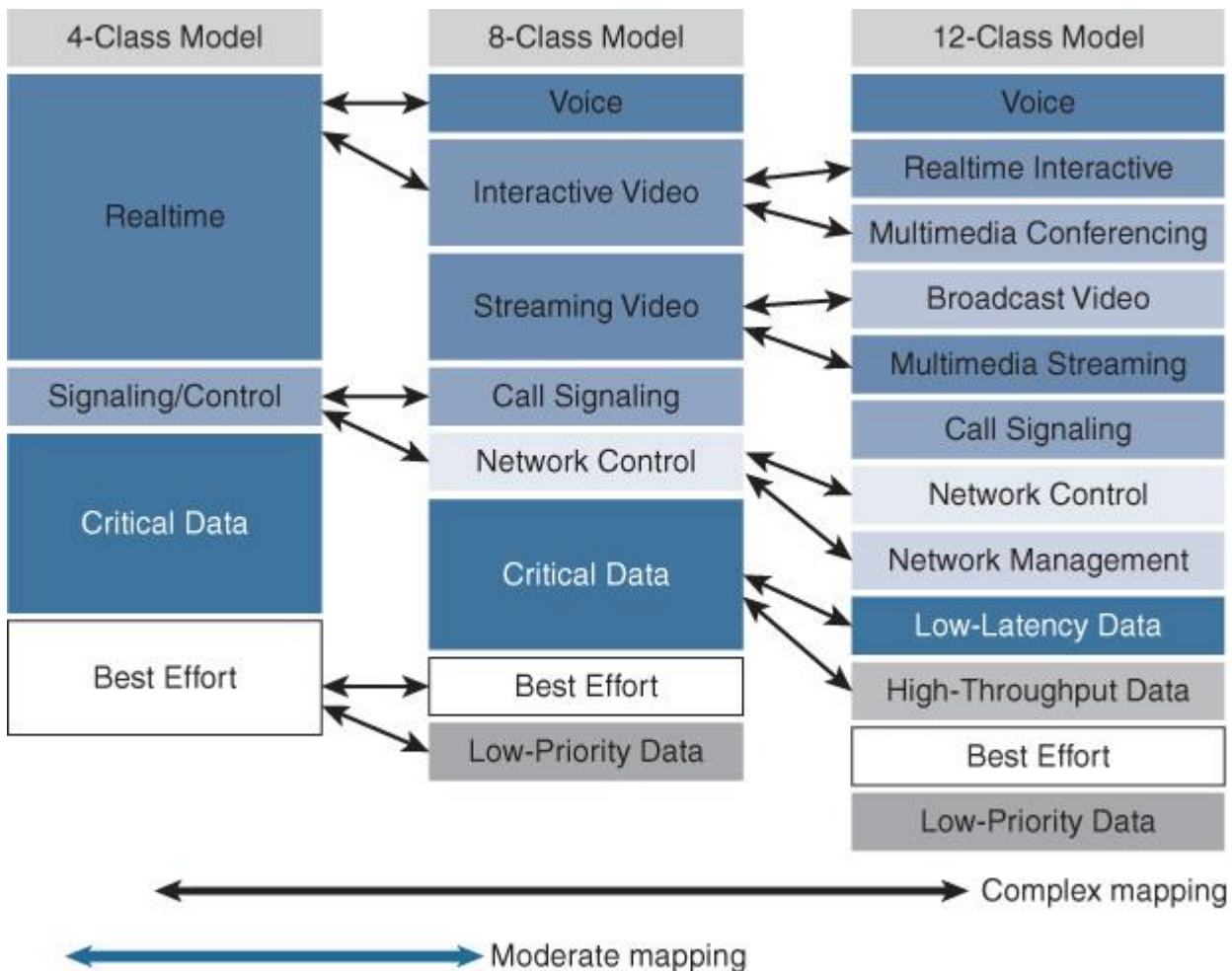


Figure 10-34 Mapping Between QoS models with Different Classes

However, both 4- and 6-class models include provisioning for only a single class for real time (usually for voice), and if video is added to the network, either a higher QoS class model (such as 8-class model) is required to be considered or both voice and video traffic have to be provisioned under a single class, which may not be a desirable option for large deployments with a large number of IP telephony and video endpoints across multiple sites. Therefore, considering the top-down approach to identifying the business and user expectations, needs, and priorities in terms of the services and applications used, in addition to their level of criticality to the business, will help, to a large extent, drive the strategy of QoS design in the right direction (business driven).

Note

Network orchestration and automation tools may help eliminate the operational complexity discussed earlier. However, this is something that depends on the configuration and change management and on the platforms and

architecture used. For instance, the level of automation in software-defined networks (SDNs) is always high, but with simplified manageability compared to other models.

Network designers must aim to balance between application requirements and QoS design with regard to the added design and operational complexity. As a general rule, network designers should always consider using the phased approach when possible, by starting with a simple QoS model (such as the 4-class model) as a baseline. In the future, if the requirements mandate more classes, it will be easier to cater to the new requirements by adding and incorporating new classes to the existing setup. This is based on the assumption that starting with a 4- or 6-class QoS model will be sufficient at that point of time with regard to the existing services and applications to be classified and profiled.

Although there are some standard recommended percentages of bandwidth allocation per class in multiple best practices design guides and white papers, such as real-time traffic (LLQ) to be assigned 33 percent of the available or interface bandwidth and the best effort class to be allocated 25 percent of the interface bandwidth, it is important to understand that these values represent a generic recommended bandwidth allocation per class based on common proven best practices. However, it is not a rule to consider these values as the standard bandwidth allocation ratios; instead, they should be considered as a foundation or a proven, tested reference when there is not enough information to build an accurate figure about bandwidth allocation in which these values can serve as a good baseline to start with. For example, [Figure 10-35](#) illustrates a classic three-tier campus network with a WAN edge. In this network, there are 20 access switches dual-homed over 1G uplinks to the distribution switches. Similarly, the distribution switches are dual-homed to the core switch over 10G links.

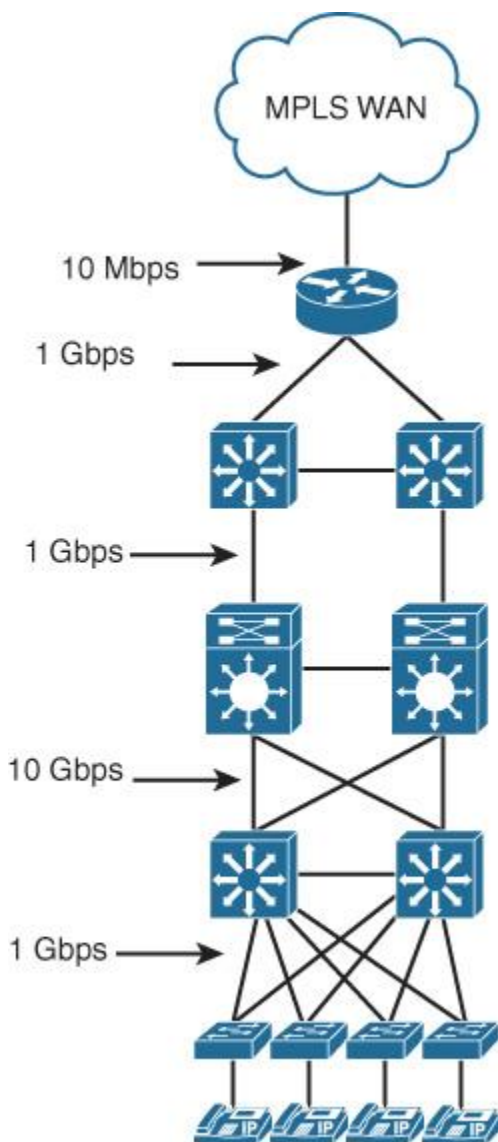


Figure 10-35 QoS Bandwidth Allocation Example

Each access switch has 30 IP phones connected to it, and VoIP calls are all based on G7.11 codec (with approximately 80 kbps required per VoIP call). Typically, with the LAN, there is no restriction on how many calls can be placed at the same time. However, over the WAN, the voice system is designed to allow a maximum of 20 simultaneous calls.

Take these requirements and map them to the standard allocation of the LLQ, which is 33 percent. Within the campus LAN, allocating 33 percent of the links across the LAN is not an issue, because the network is already overprovisioned. However, provisioning the real-time (VoIP) LLQ with 33 percent of the 10G uplinks means that there will be 3.3G of bandwidth reserved during a period of congestion. This can be seen as a security risk by the security team, because someone can flood the network with malicious traffic marked as DSCP EF, which will take precedence over other legitimate traffic and will consume the available 3.3G of bandwidth per uplink.

The WAN edge, however, has a 10-Mbps WAN link. Based on the voice system design, there will be only 20 simultaneous calls, meaning a maximum of (80 kbps * 20) 1.6 Mbps. Based on this, allocating 33 percent (= 3.3M) of the available WAN interface bandwidth to the LLQ (VoIP) means a waste of bandwidth in this case. Nevertheless, in this scenario, this organization still can allocate 25 percent of the links' available bandwidth to best effort (the default class) because there is no absolute requirements here on that part. Therefore, network designers need to consider the best practice recommended bandwidth or percentage allocation per class as a baseline and tweak the allocation based on the actual traffic flow requirements, taking into consideration security concerns, the available amount of bandwidth, and whether it is across the WAN, LAN, or a data center network.

Taking into consideration the different QoS design aspects, techniques, and strategies covered earlier in this section, network designers can consider the QoS design framework shown in [Figure 10-36](#) as a foundational reference when approaching any QoS design task during the planning phase.

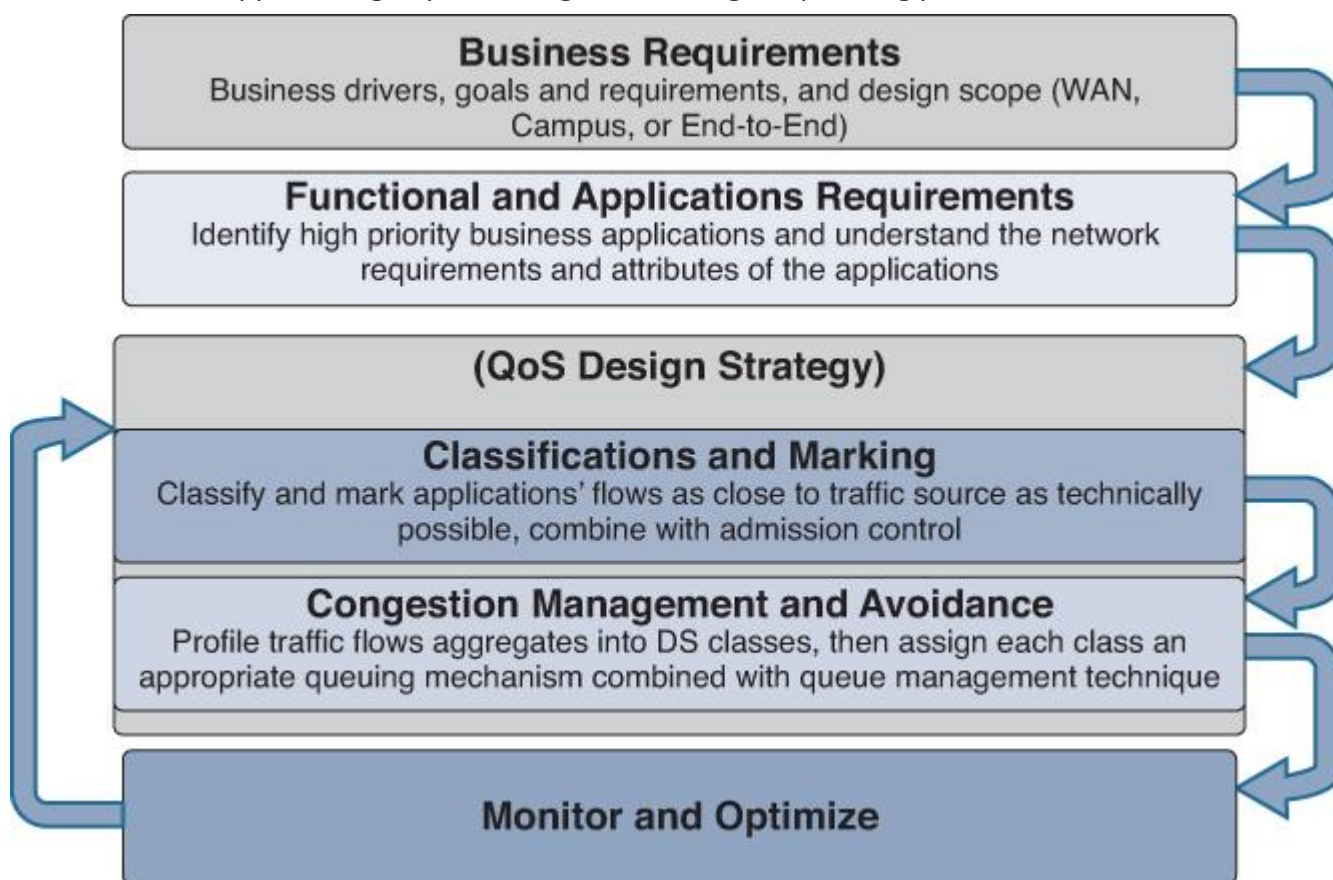


Figure 10-36 QoS Design Framework

Enterprise QoS Design Considerations

This section discusses QoS considerations that are applicable to enterprise-grade networks.